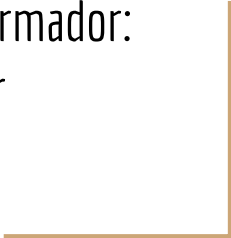




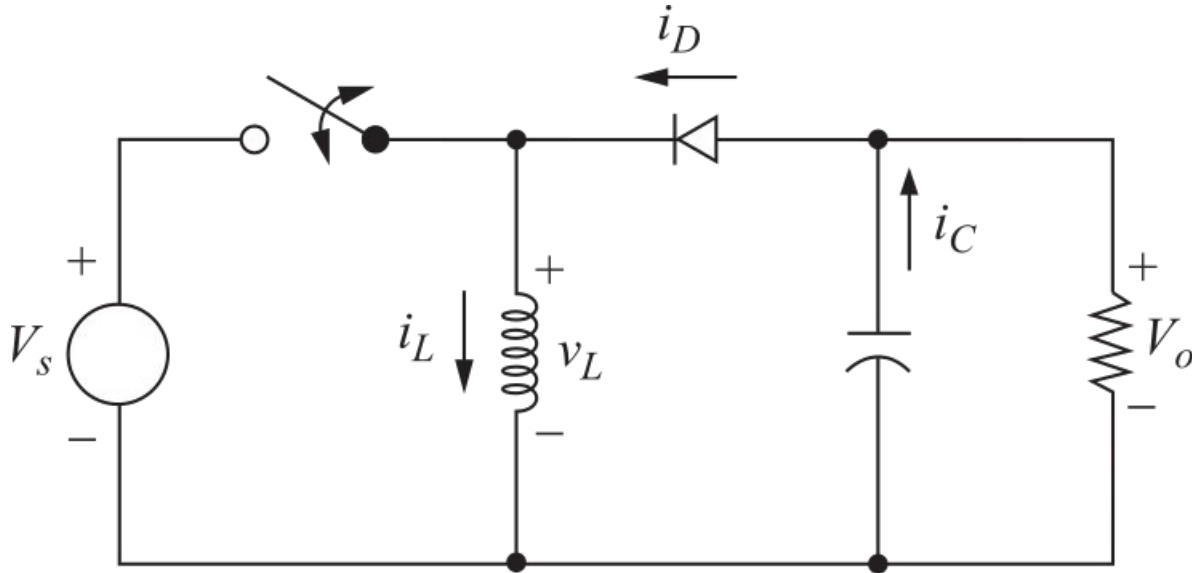
Conversión CC/CC

Convertidores sin Transformador:
Reductor-Elevador

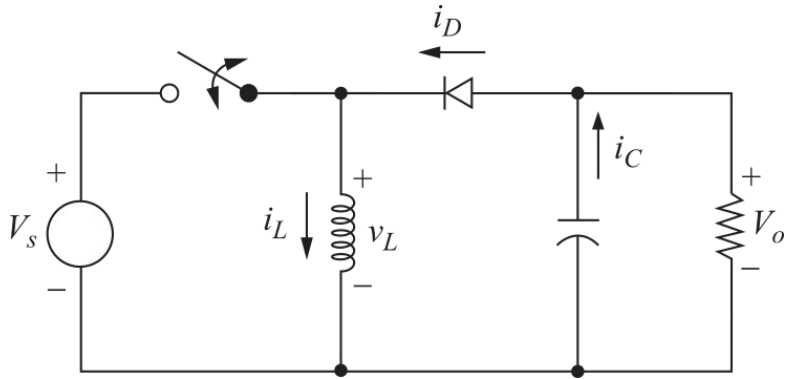


Convertidor Reductor-Elevador: circuito

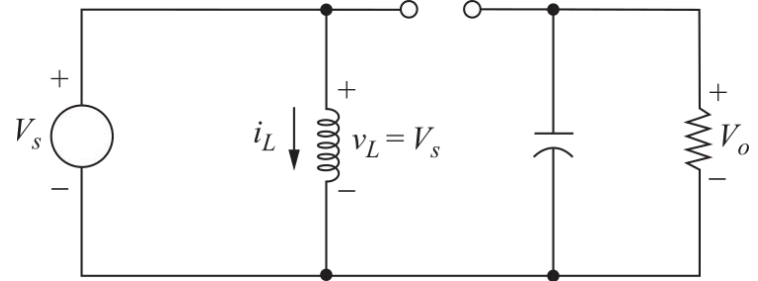
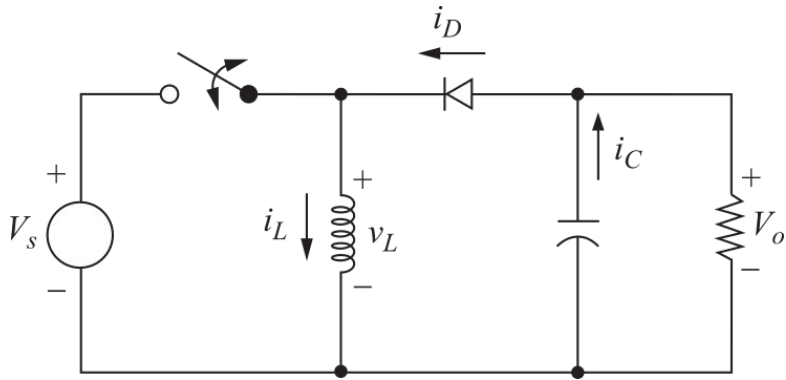
El convertidor Reductor-Elevador tiene la capacidad de generar una magnitud de tensión de salida mayor o menor que la tensión de entrada, en función del ciclo de trabajo D . Utiliza los mismos componentes, conectados de forma diferente.



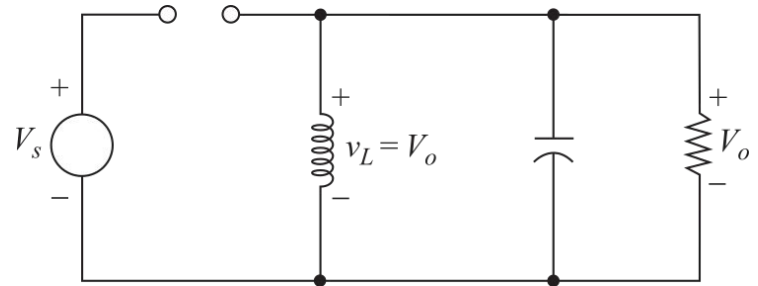
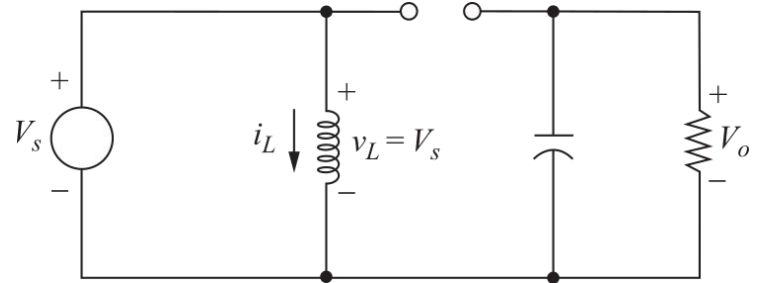
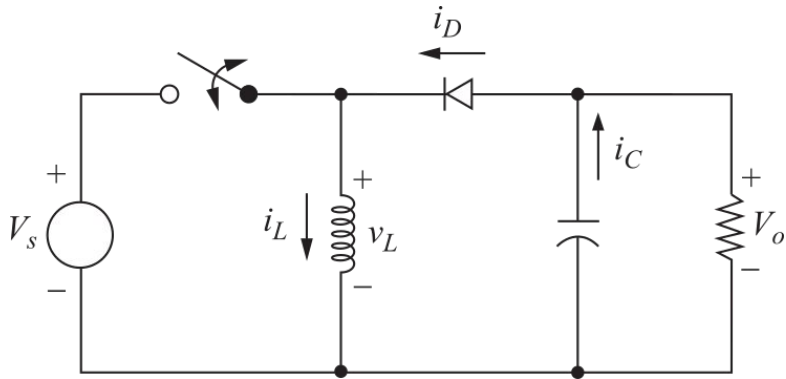
Convertidor Reductor-Elevador: funcionamiento



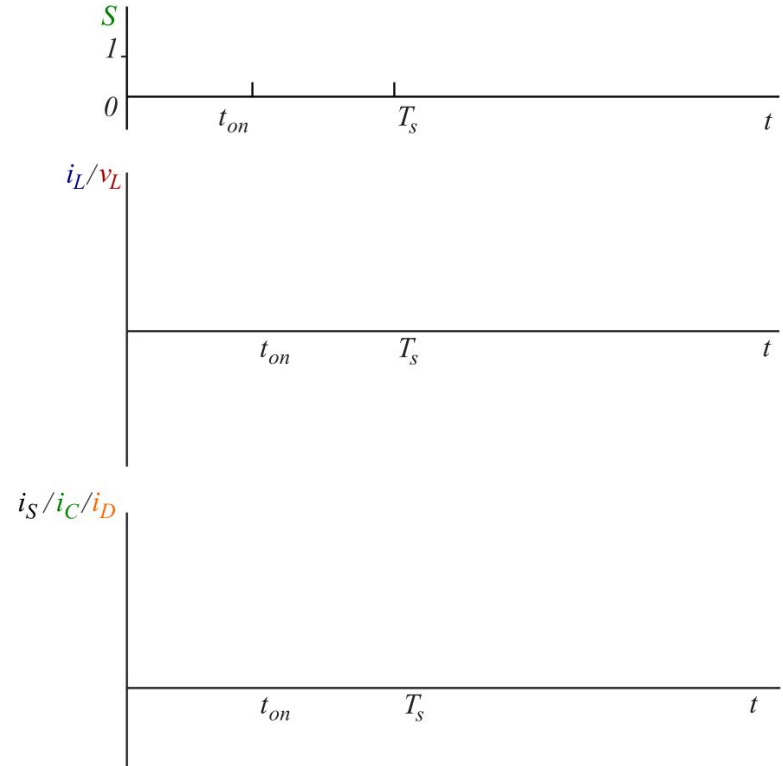
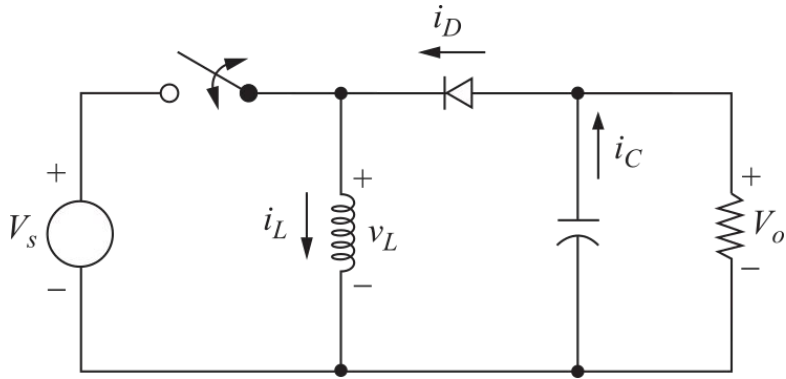
Convertidor Reductor-Elevador: funcionamiento



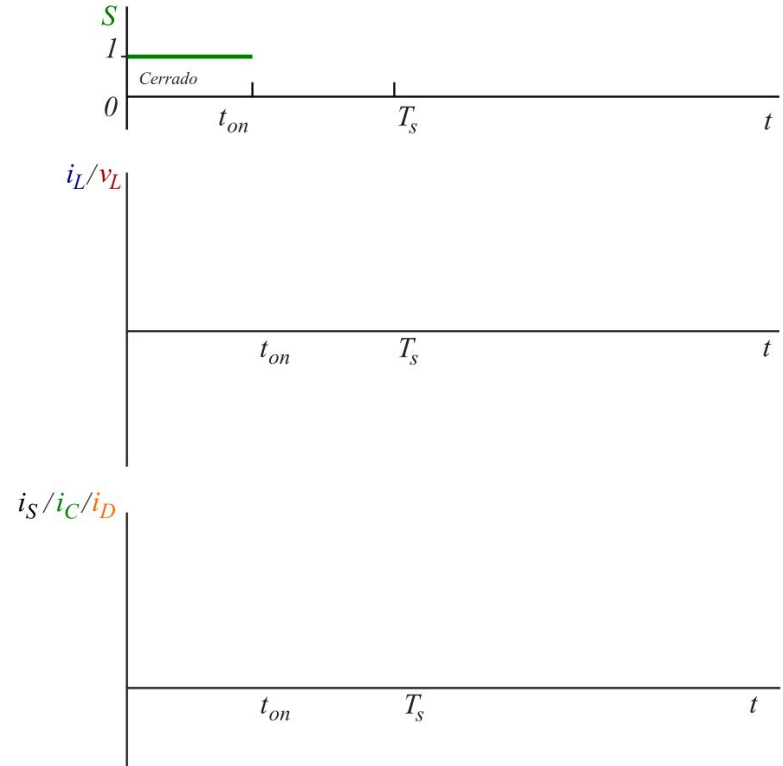
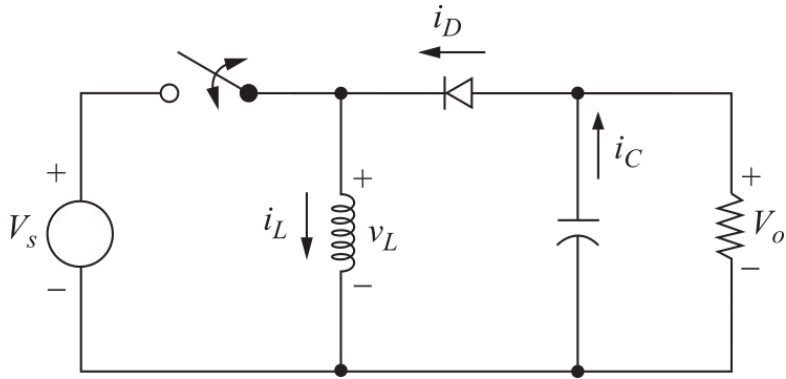
Convertidor Reductor-Elevador: funcionamiento



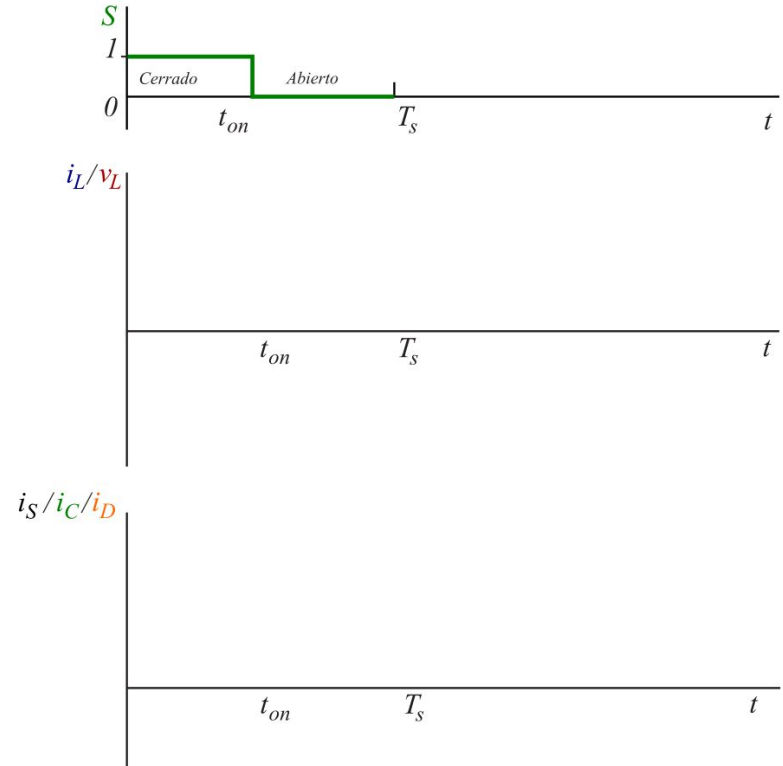
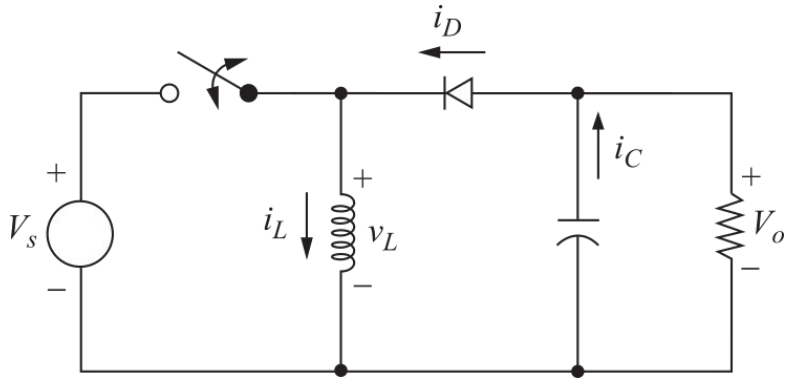
Convertidor Reductor-Elevador: formas de onda



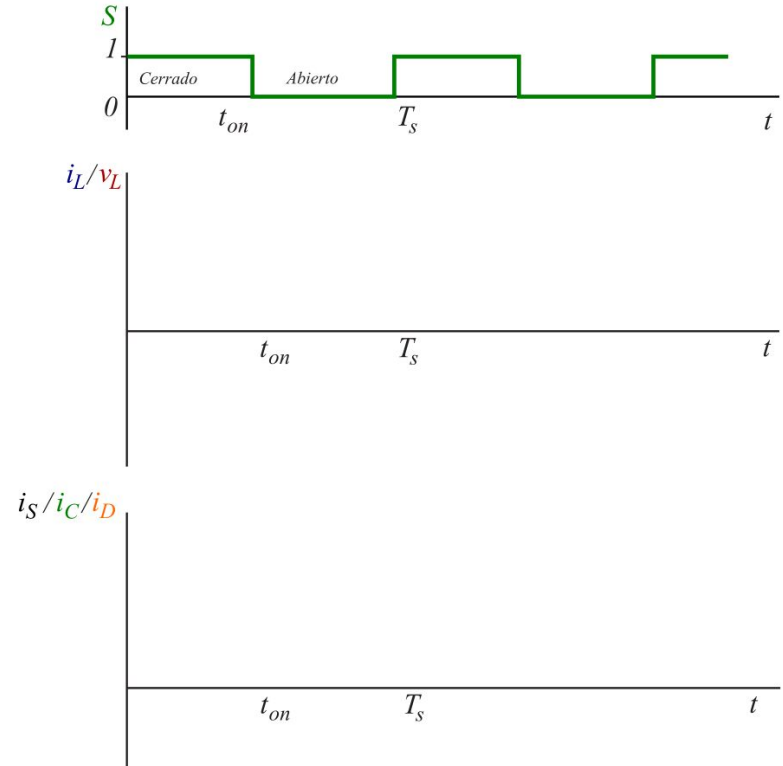
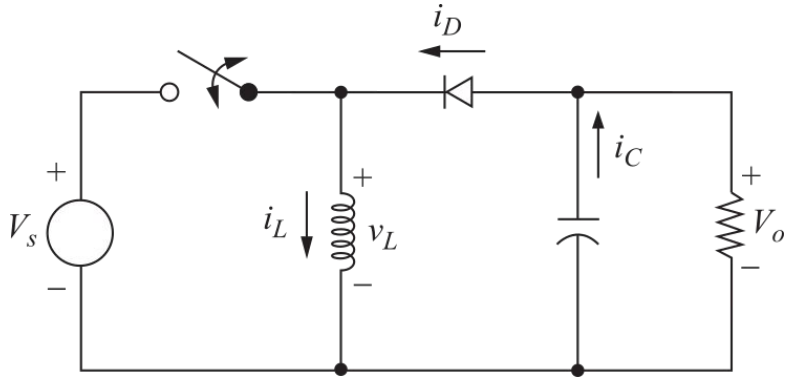
Convertidor Reductor-Elevador: formas de onda



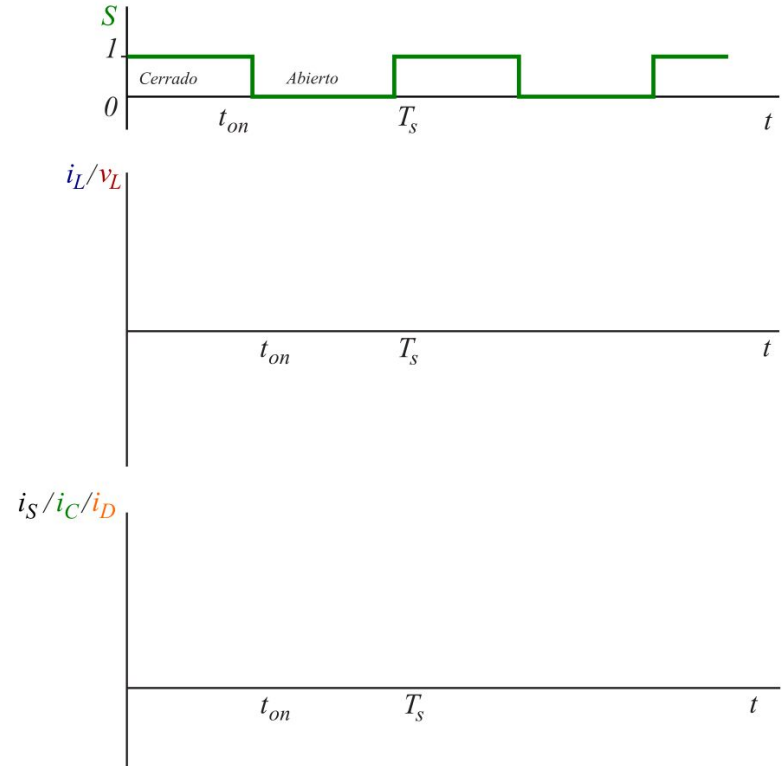
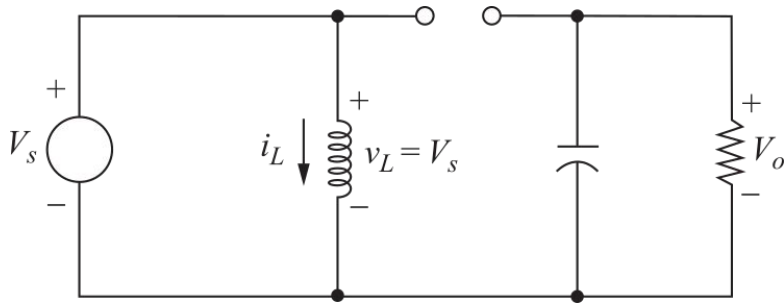
Convertidor Reductor-Elevador: formas de onda



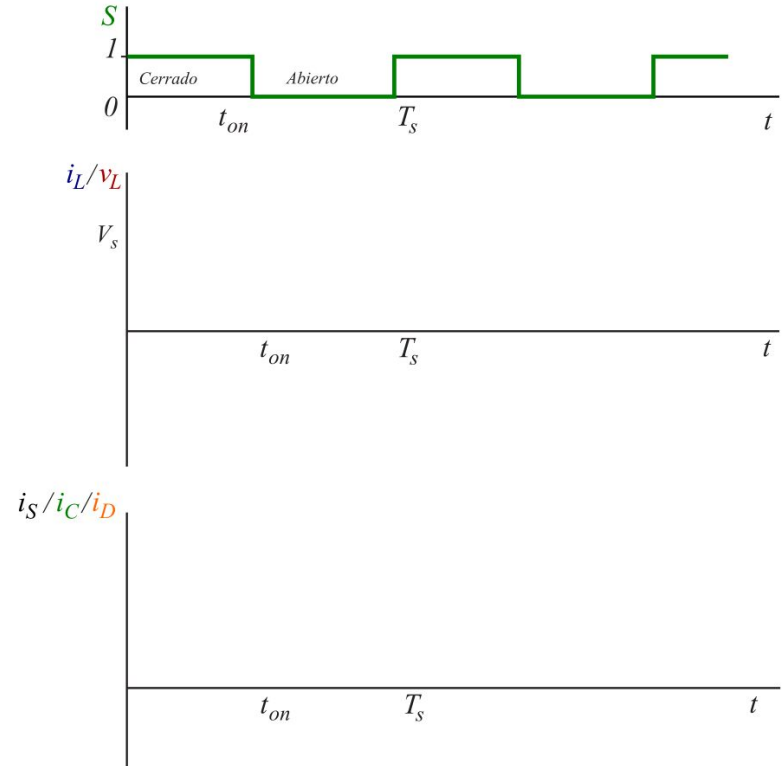
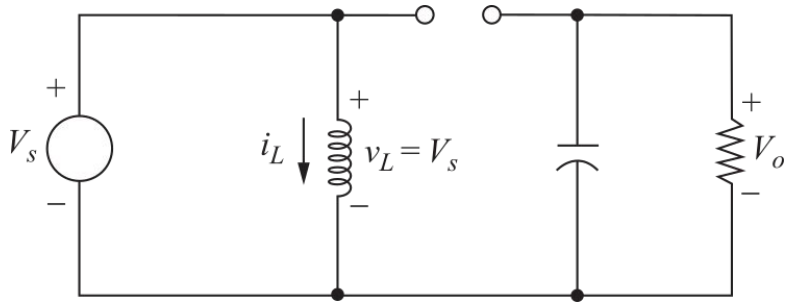
Convertidor Reductor-Elevador: formas de onda



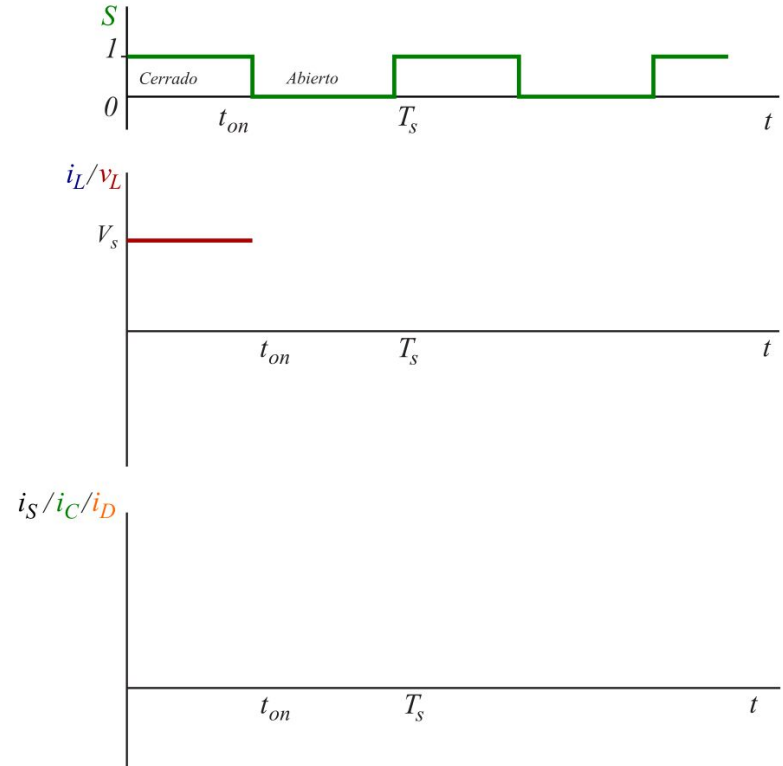
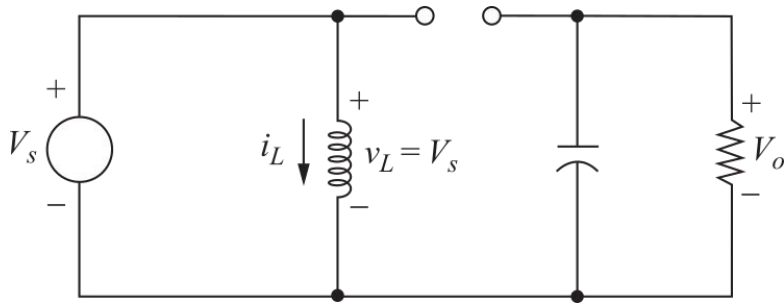
Convertidor Reductor-Elevador: formas de onda



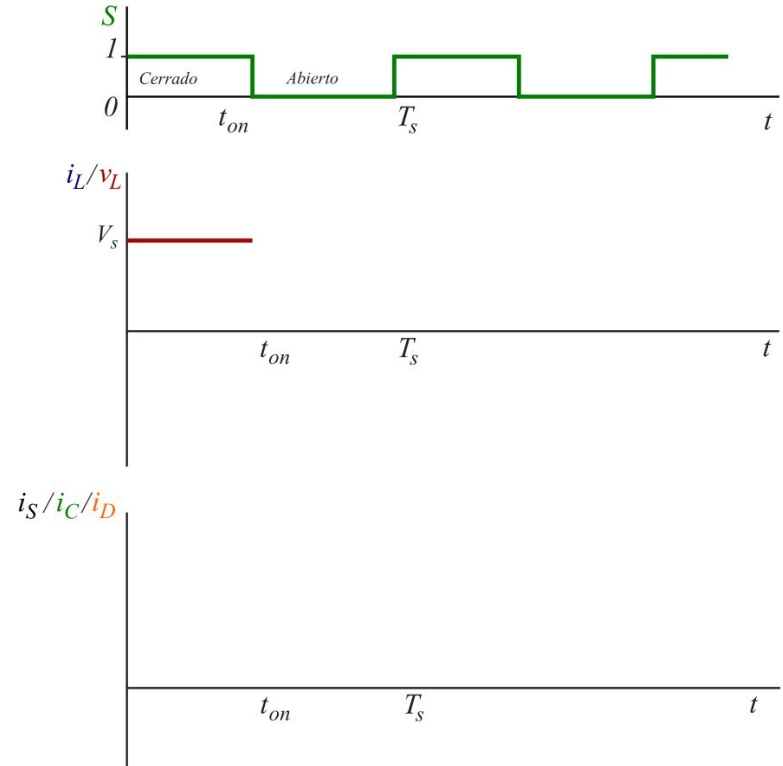
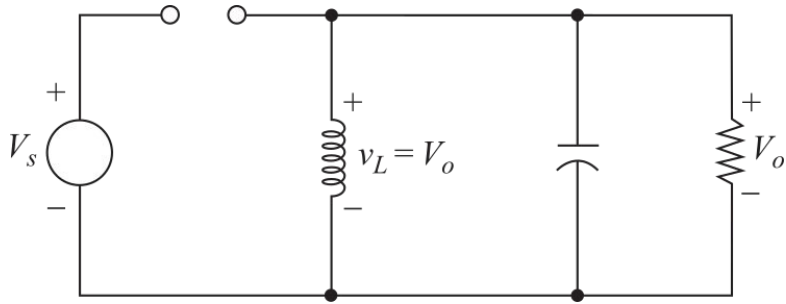
Convertidor Reductor-Elevador: formas de onda



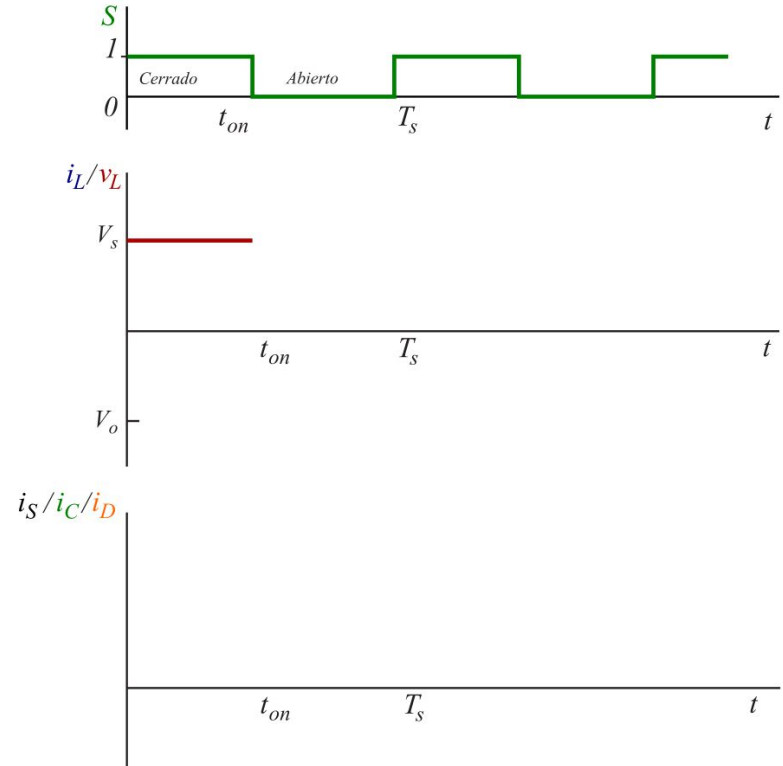
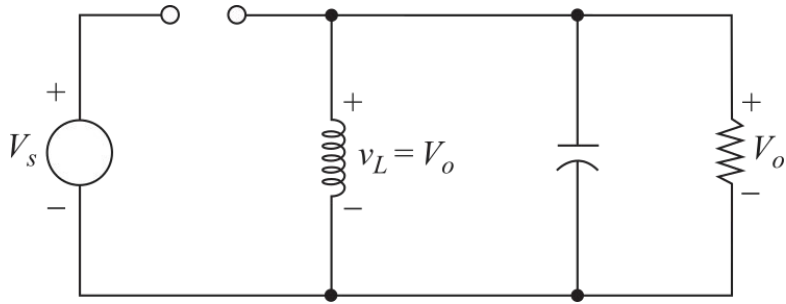
Convertidor Reductor-Elevador: formas de onda



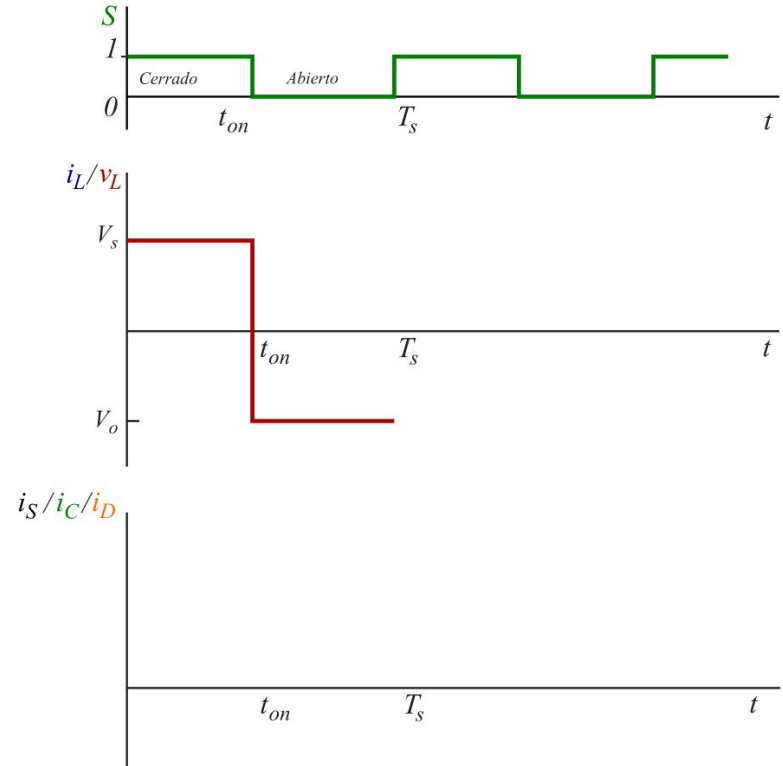
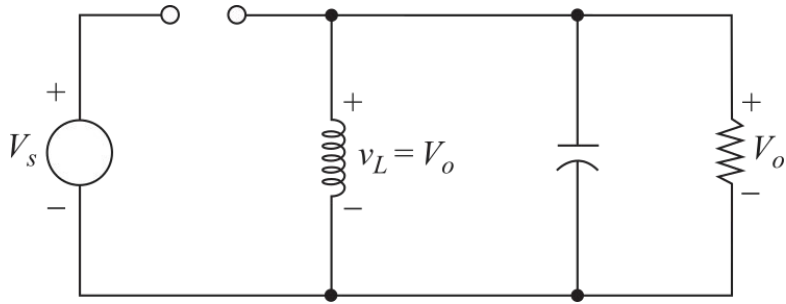
Convertidor Reductor-Elevador: formas de onda



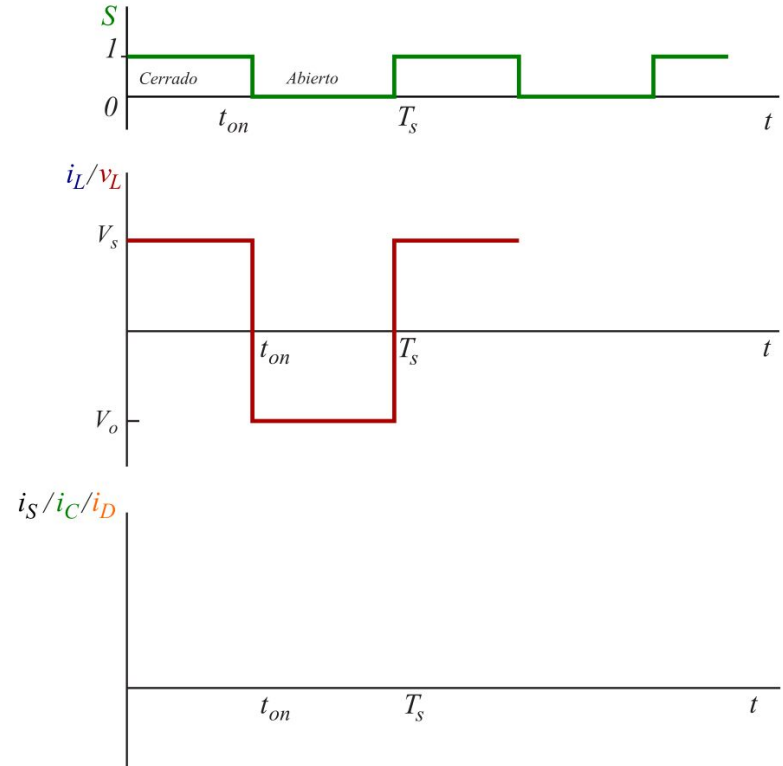
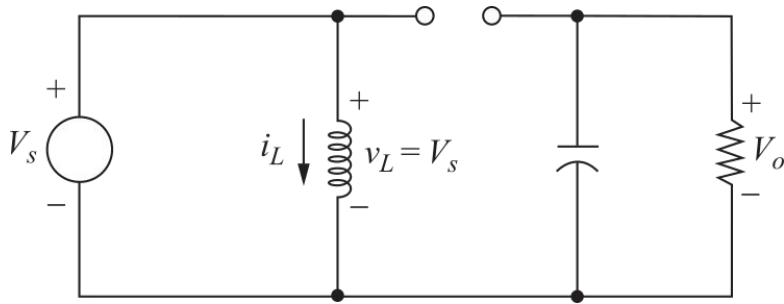
Convertidor Reductor-Elevador: formas de onda



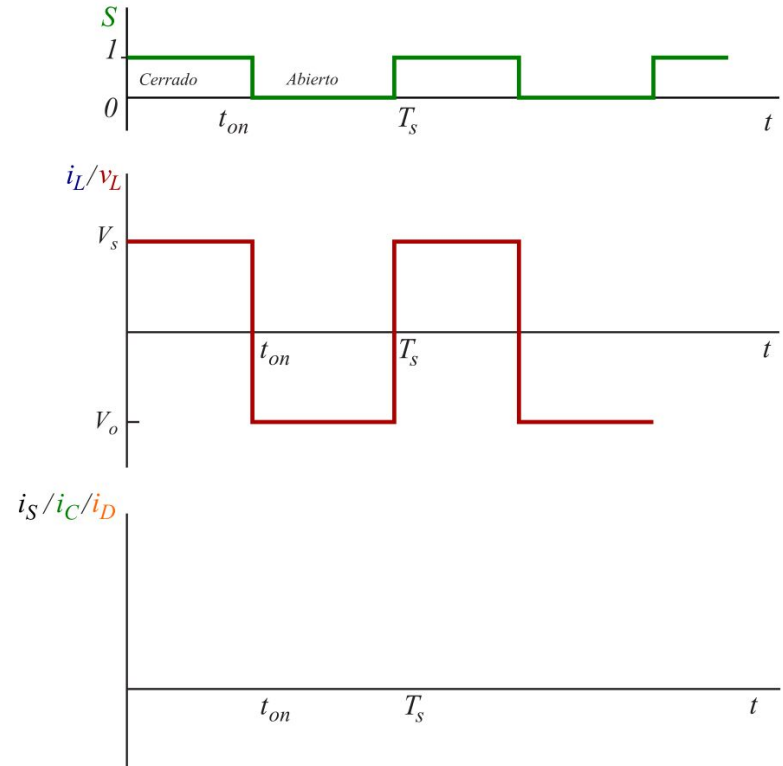
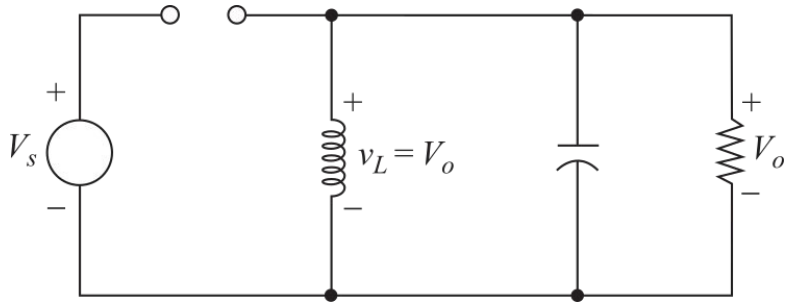
Convertidor Reductor-Elevador: formas de onda



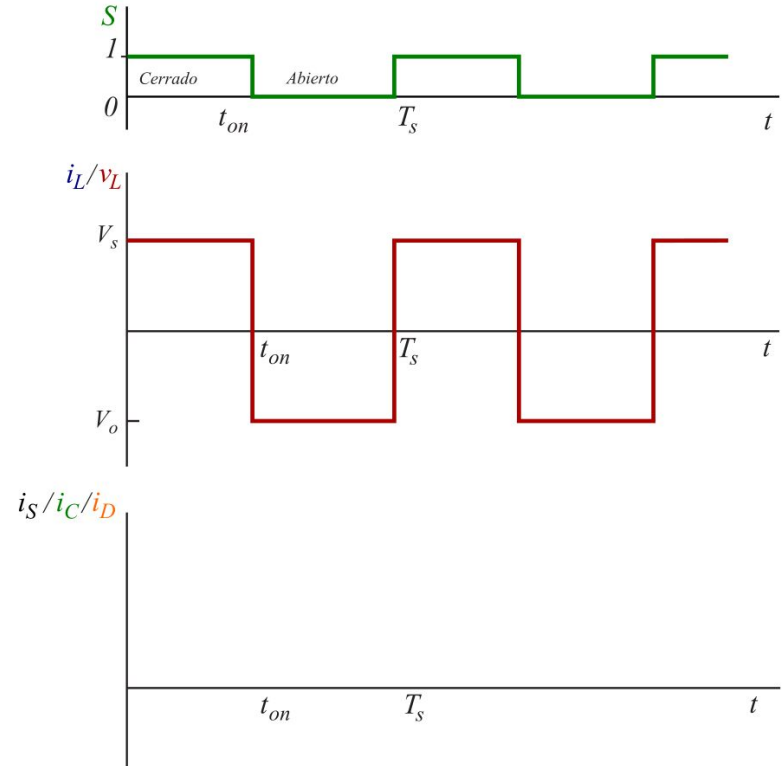
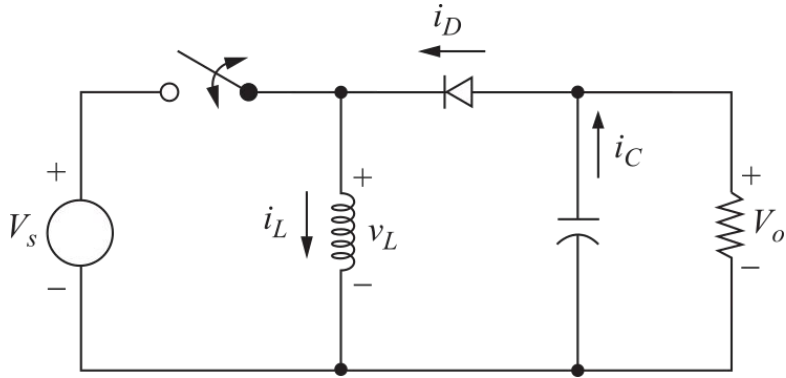
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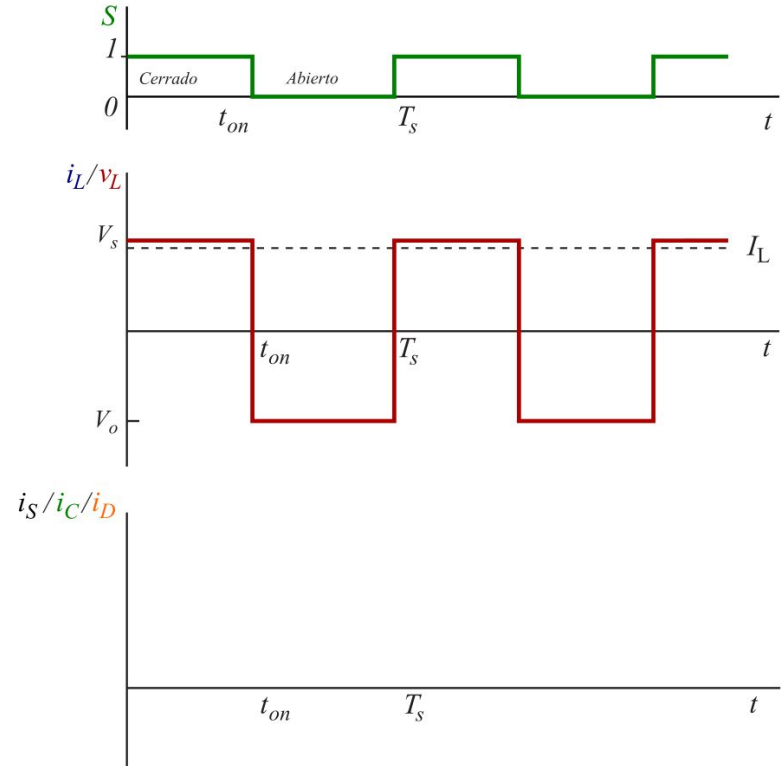
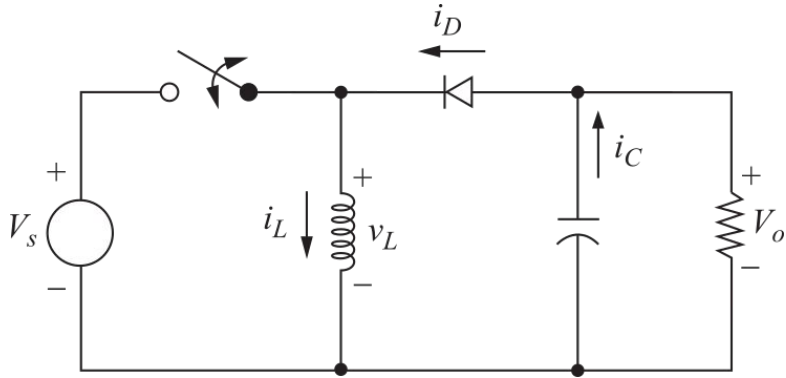
Convertidor Reductor-Elevador: formas de onda



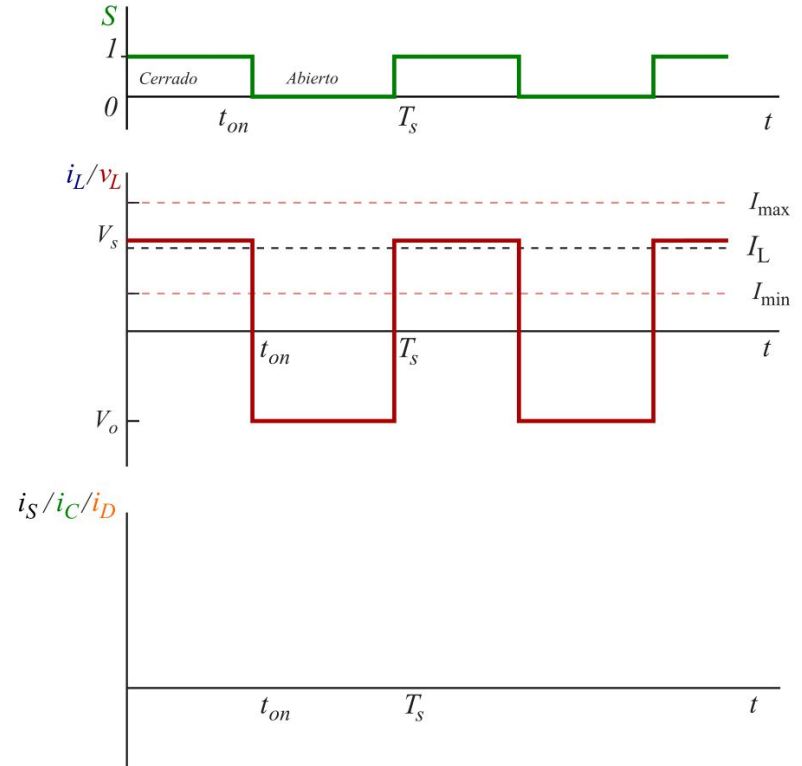
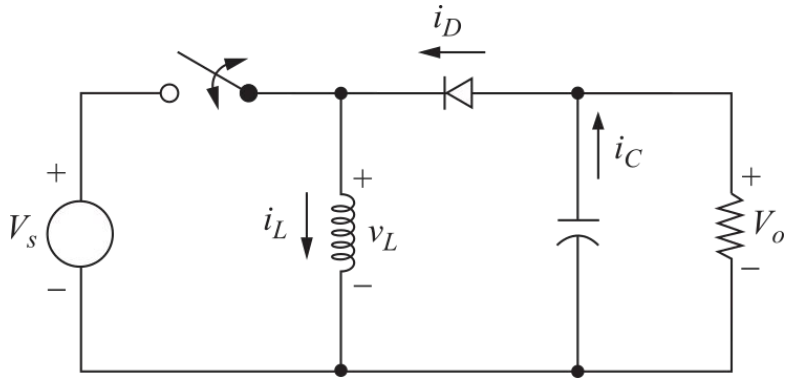
Convertidor Reductor-Elevador: formas de onda



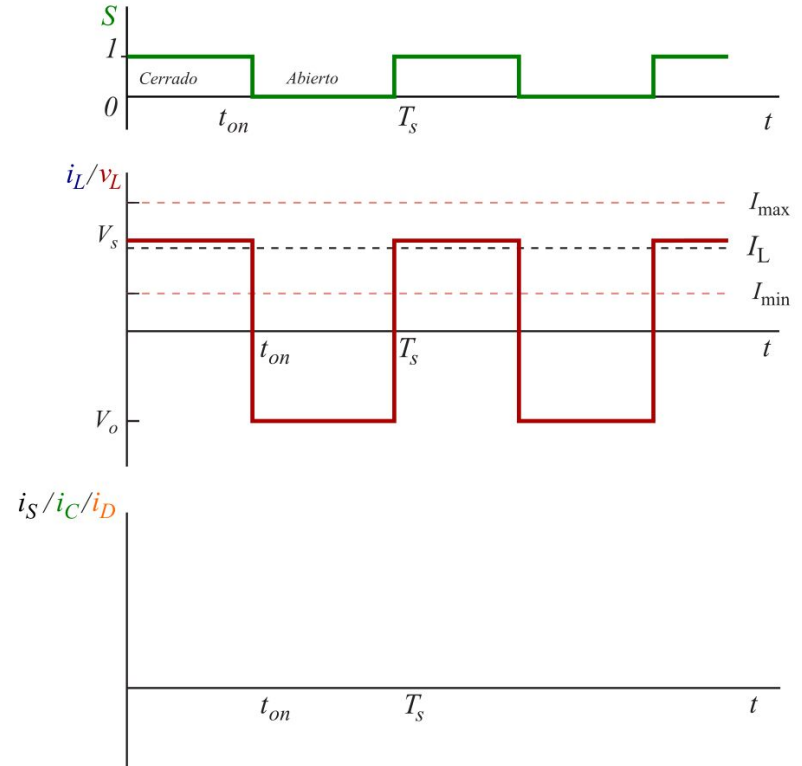
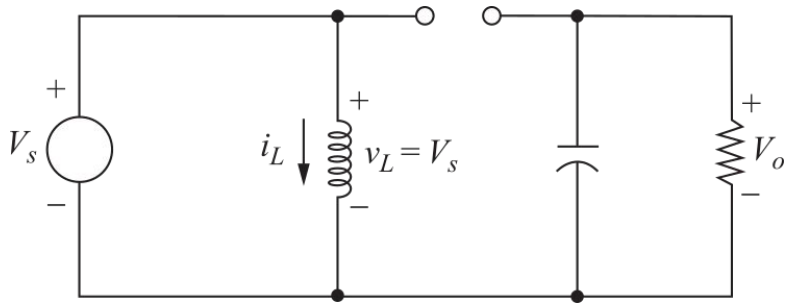
Convertidor Reductor-Elevador: formas de onda



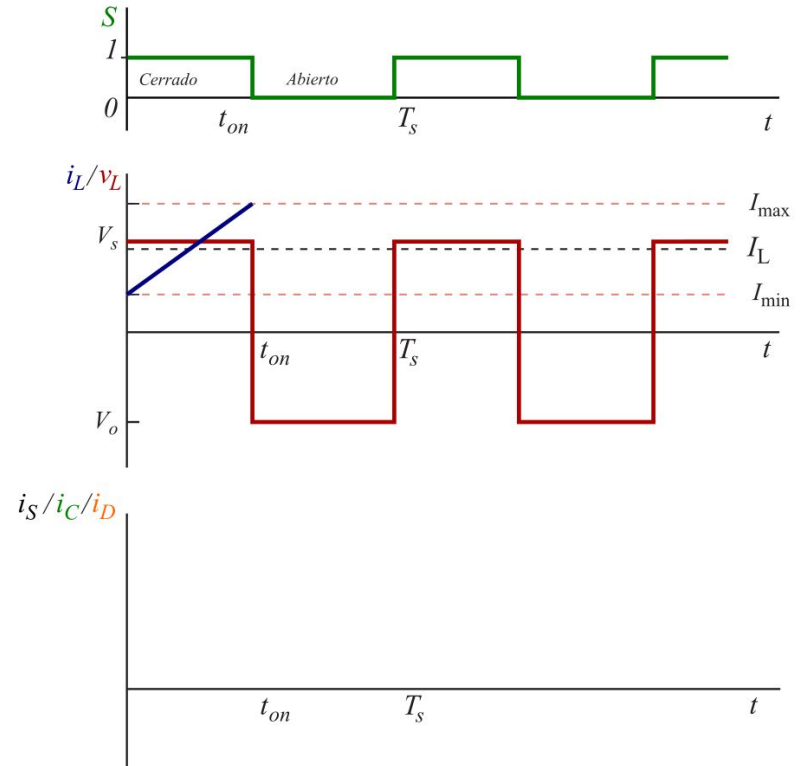
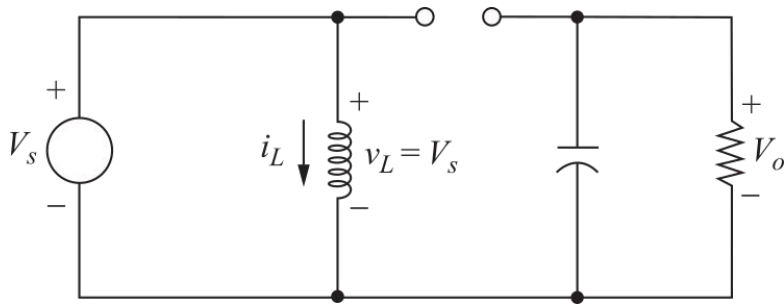
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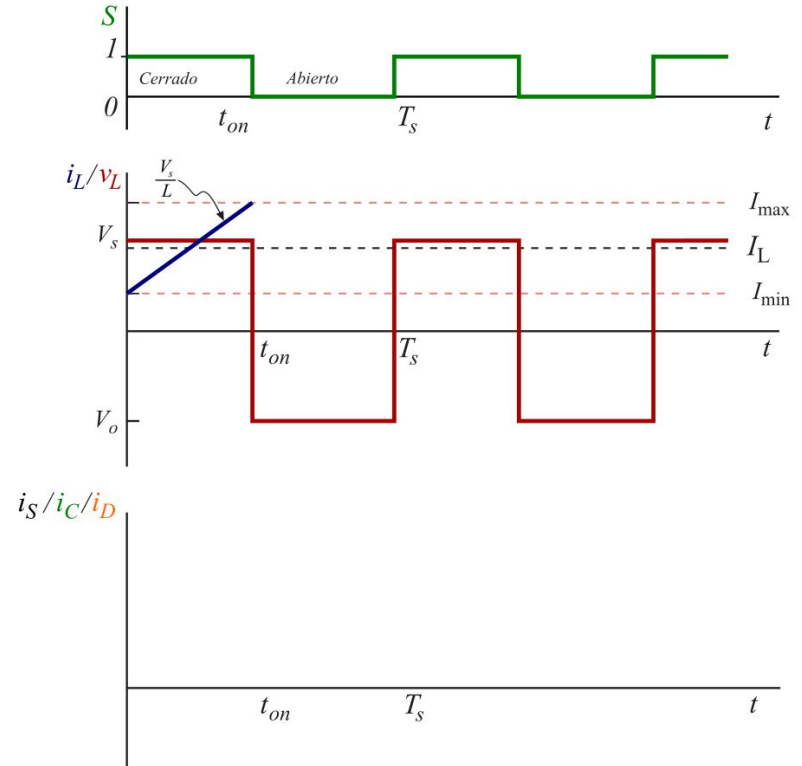
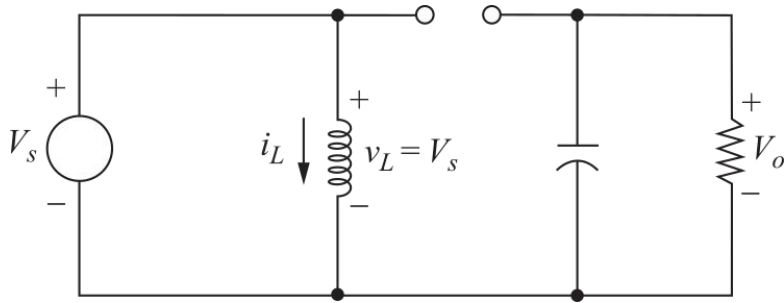
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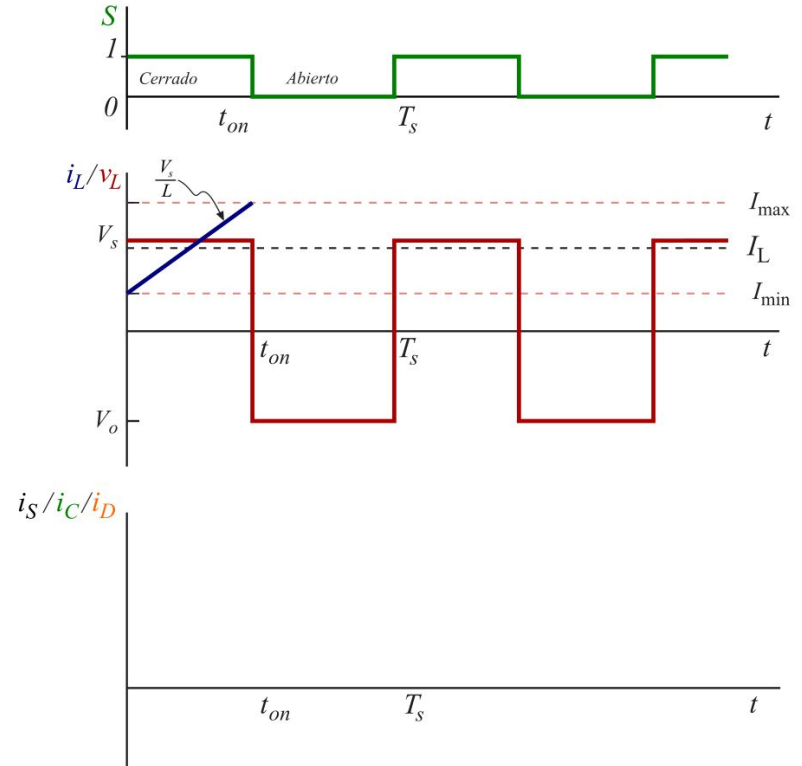
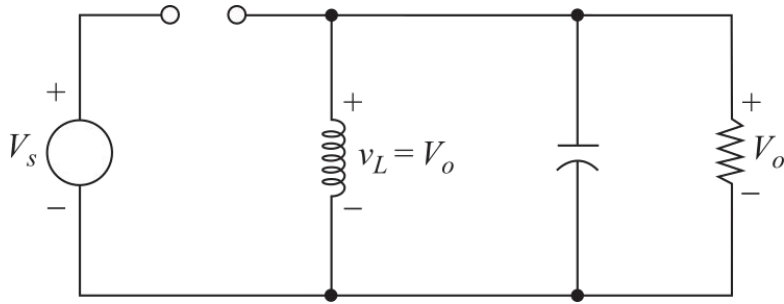
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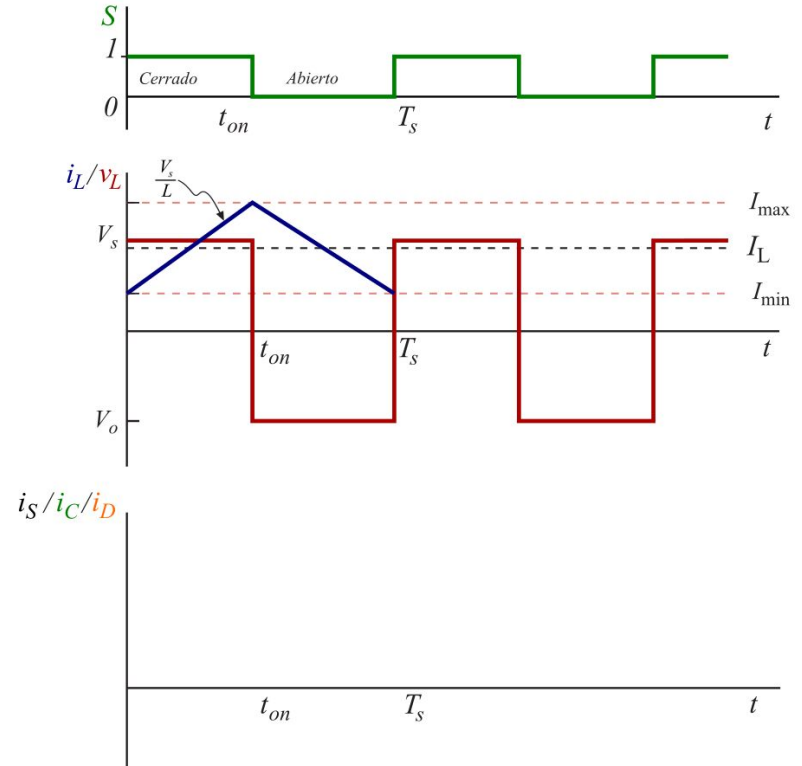
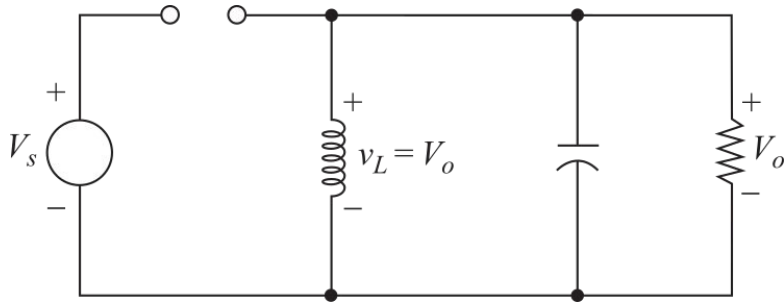
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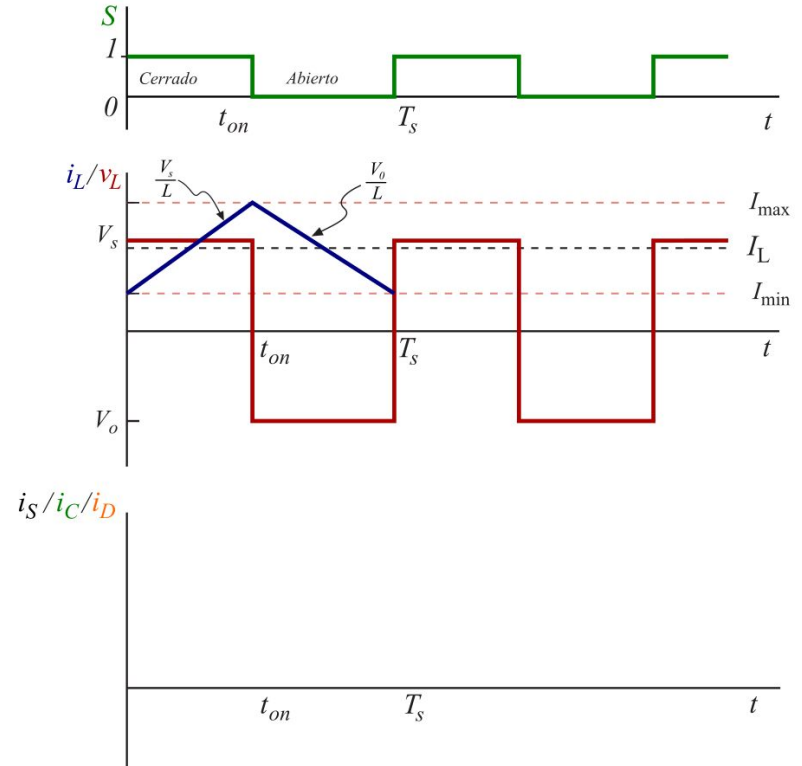
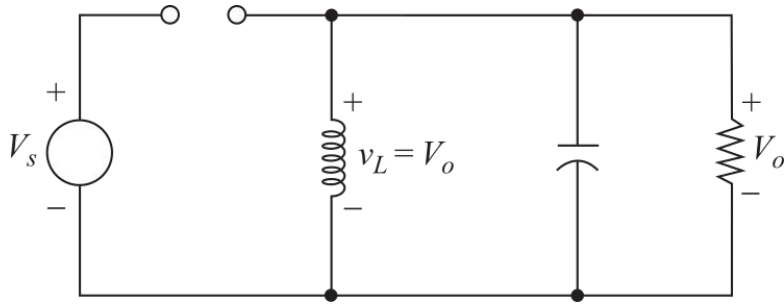
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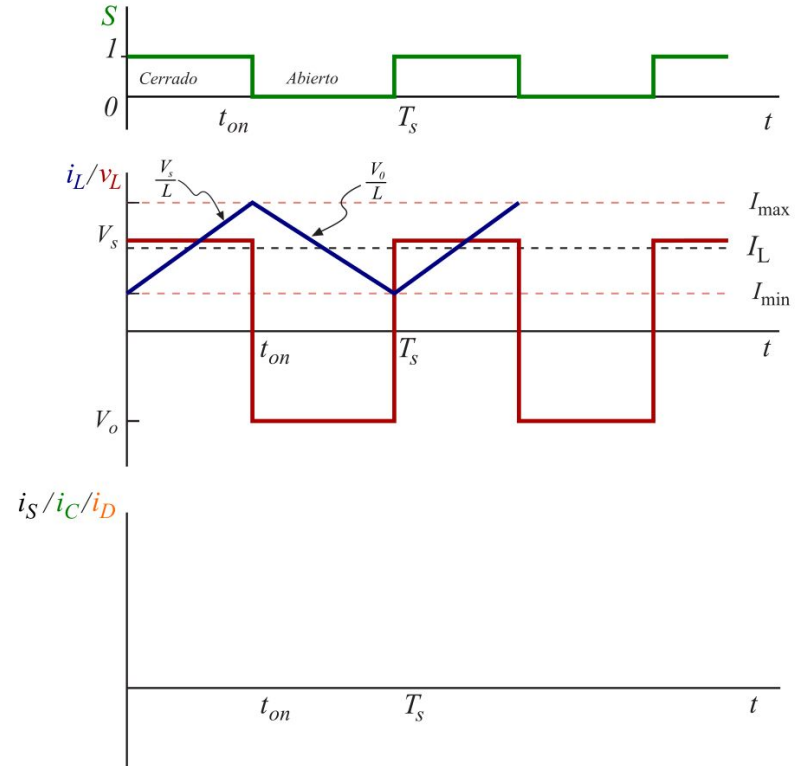
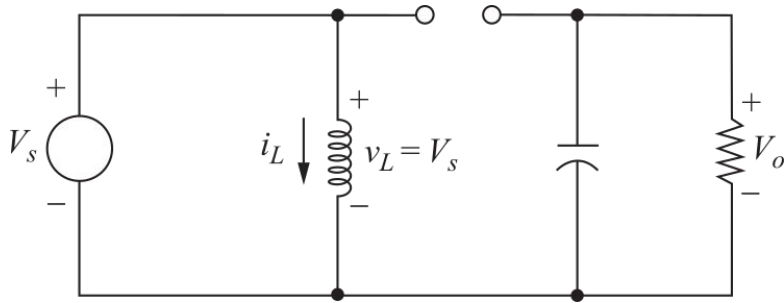
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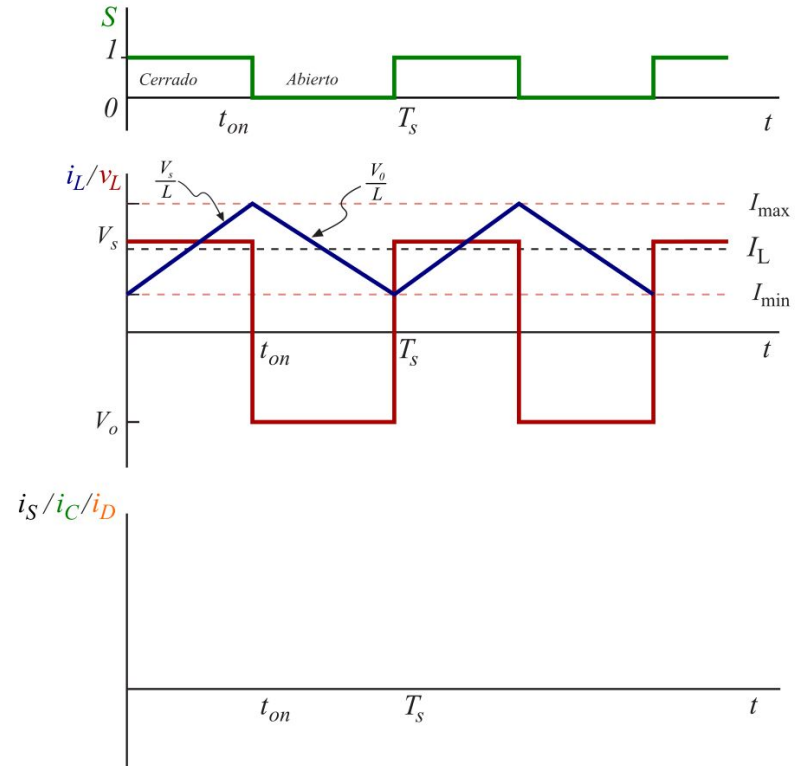
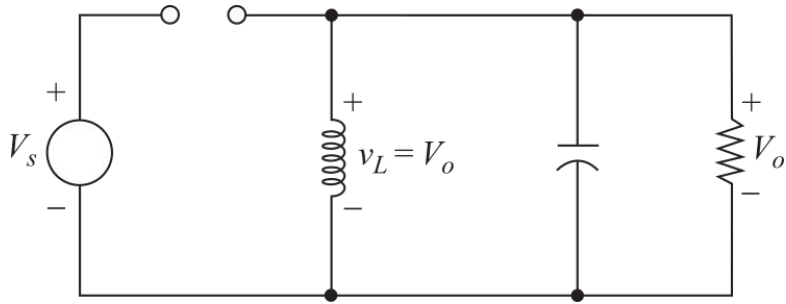
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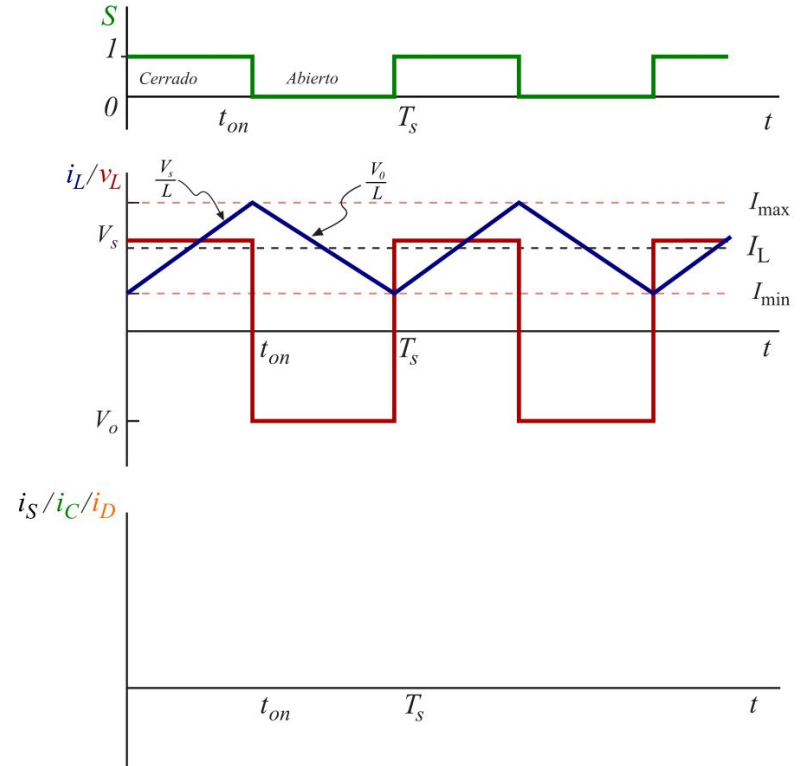
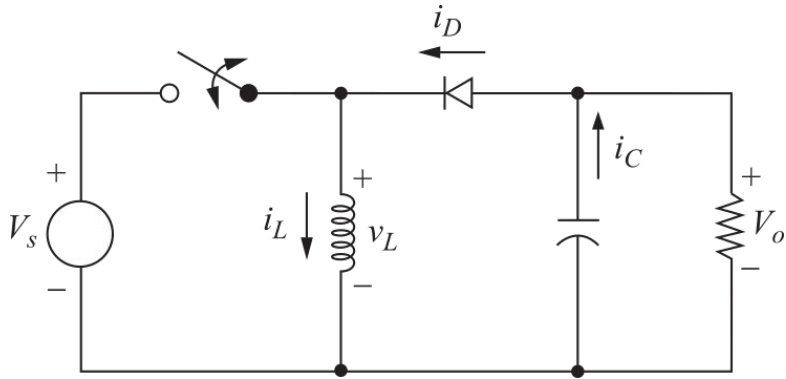
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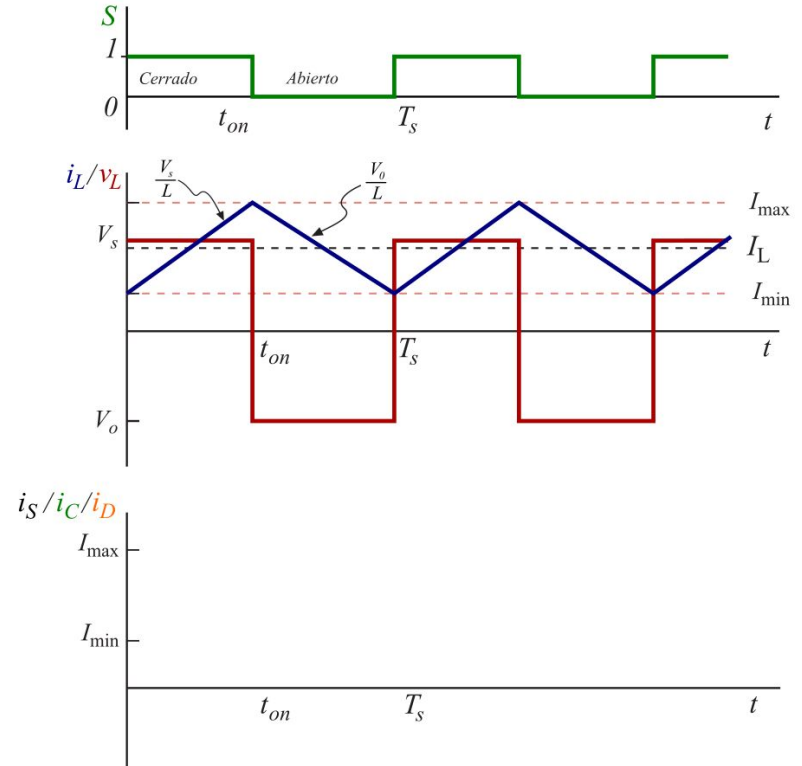
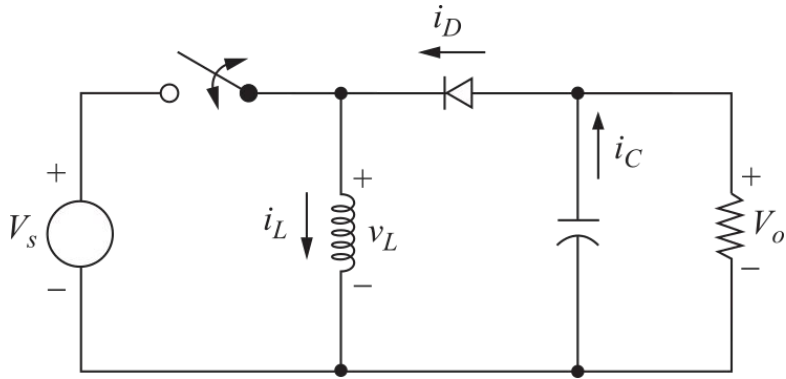
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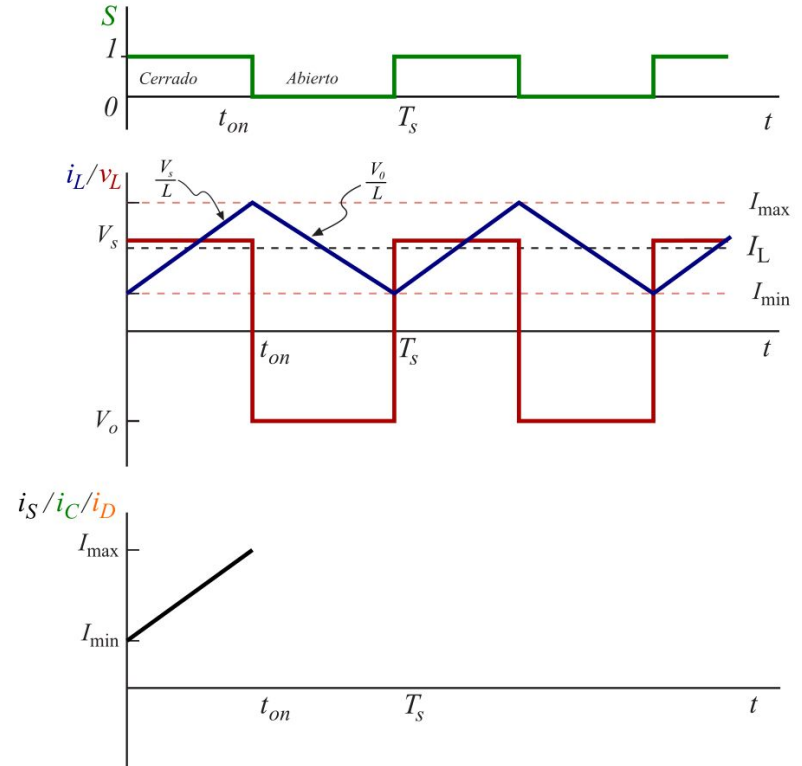
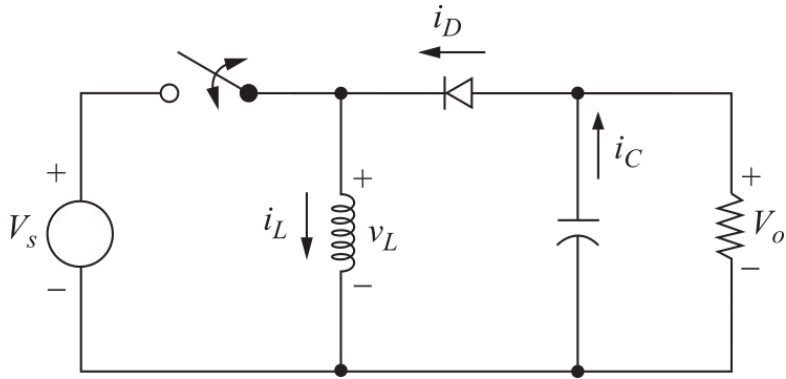
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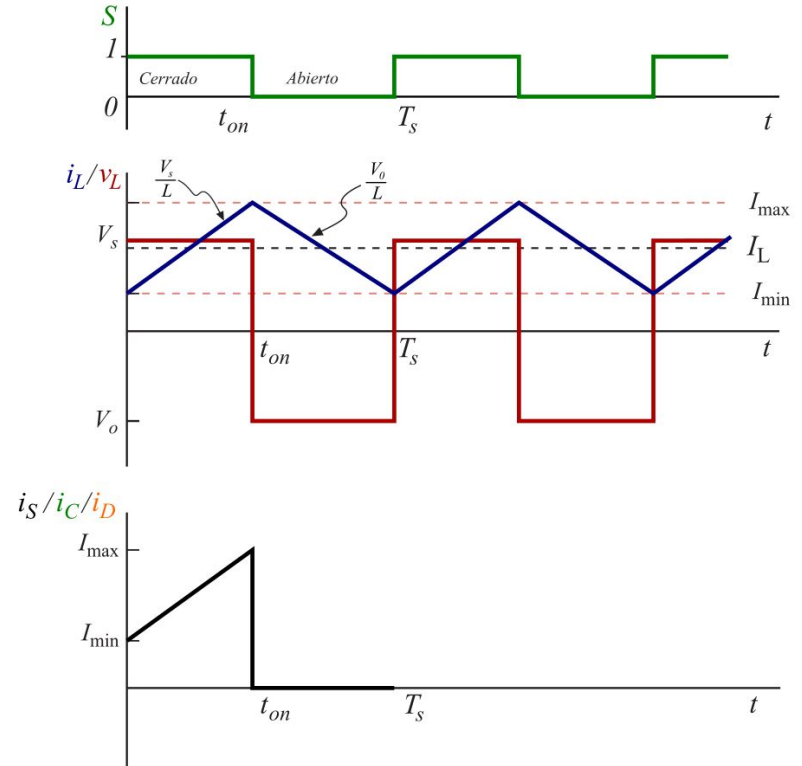
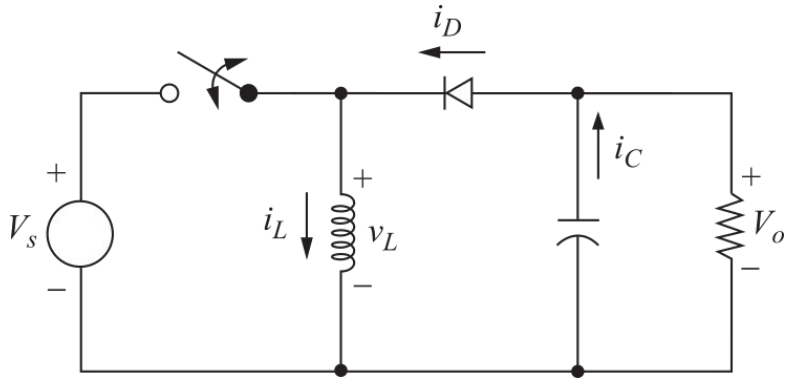
Convertidor Reductor-Elevador: formas de onda



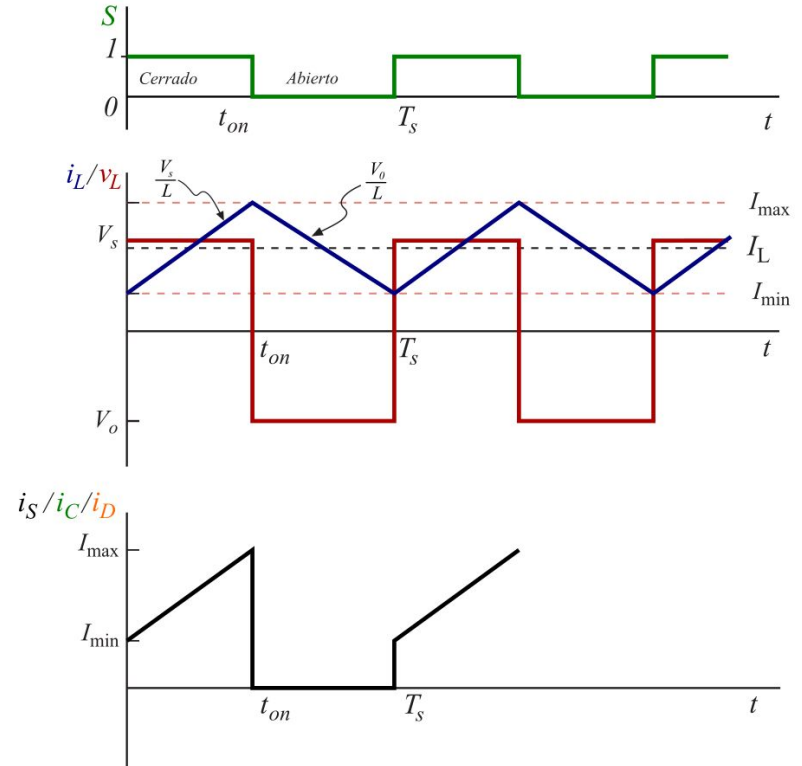
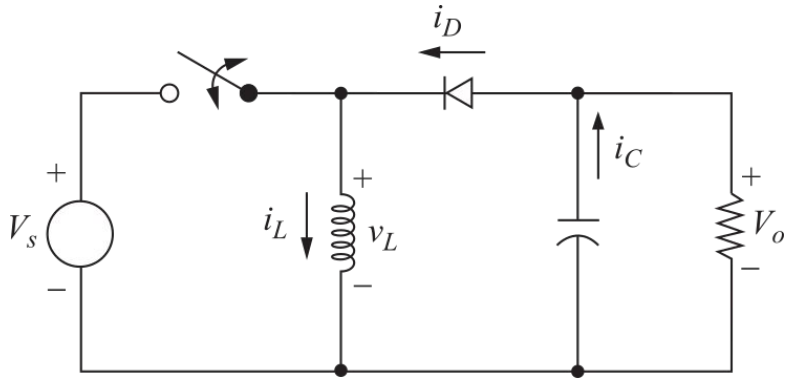
Convertidor Reductor-Elevador: formas de onda



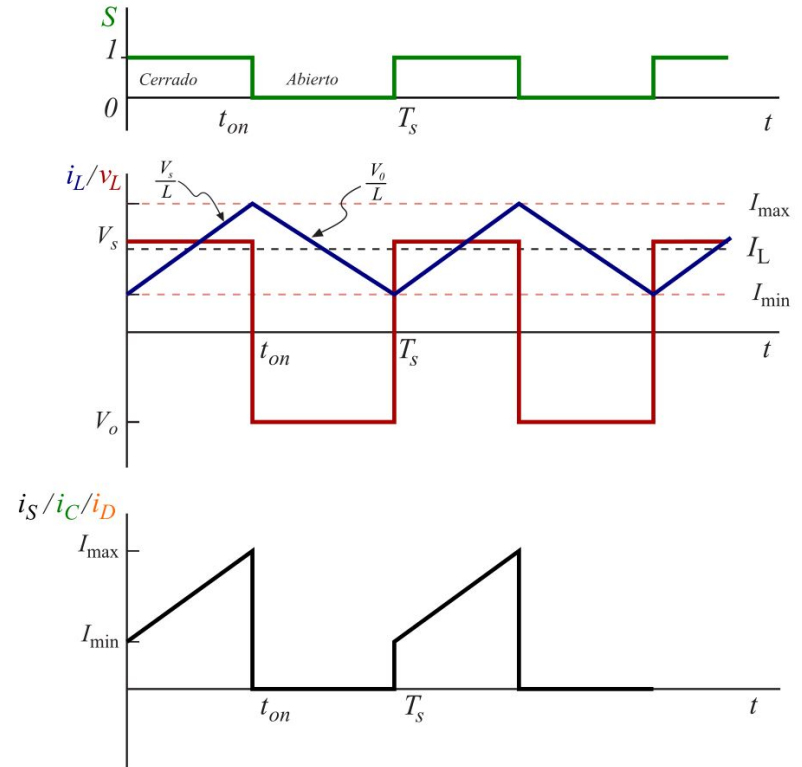
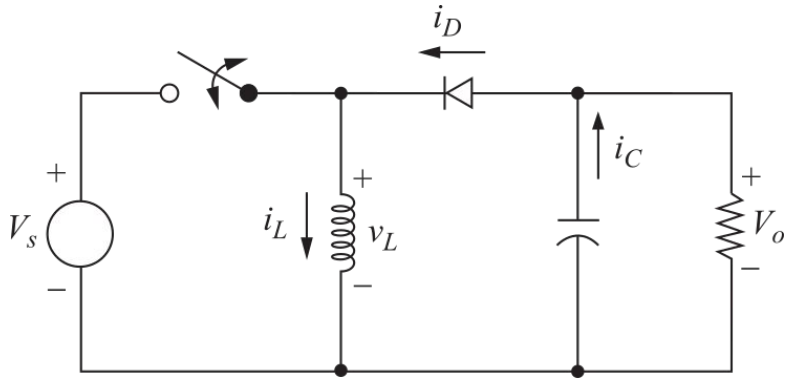
Convertidor Reductor-Elevador: formas de onda



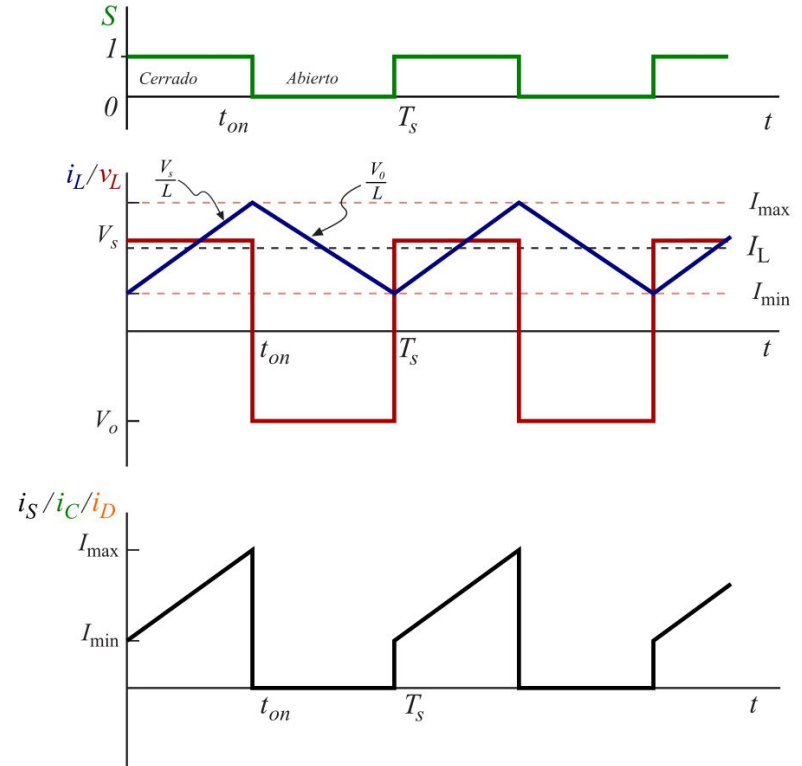
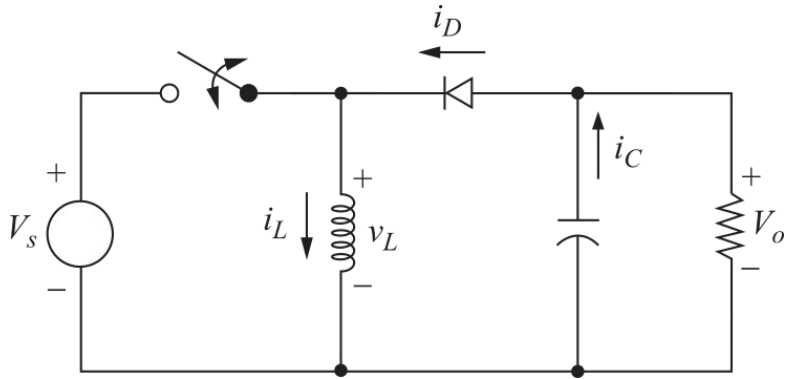
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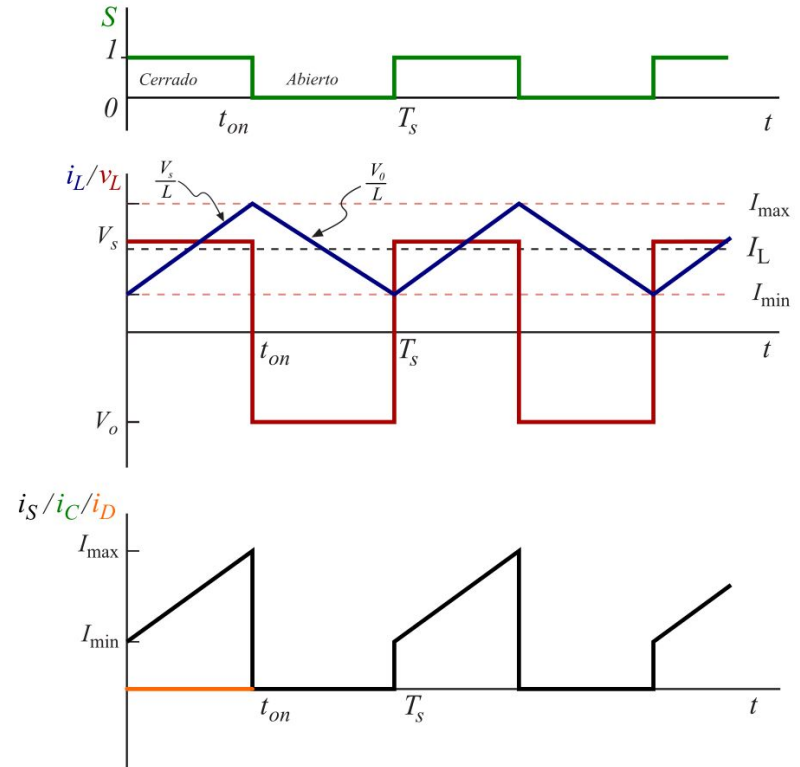
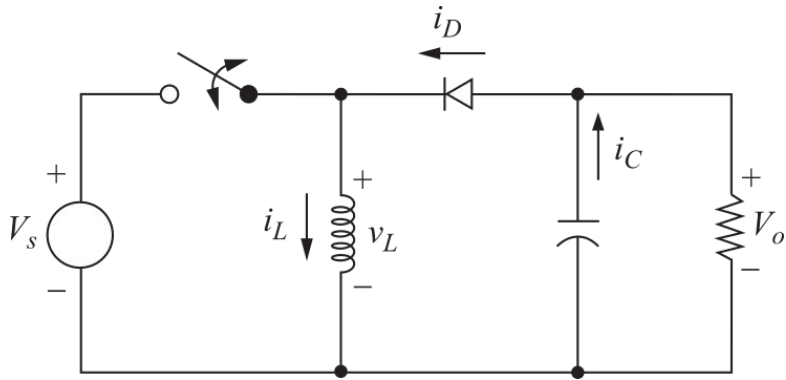
Convertidor Reductor-Elevador: formas de onda



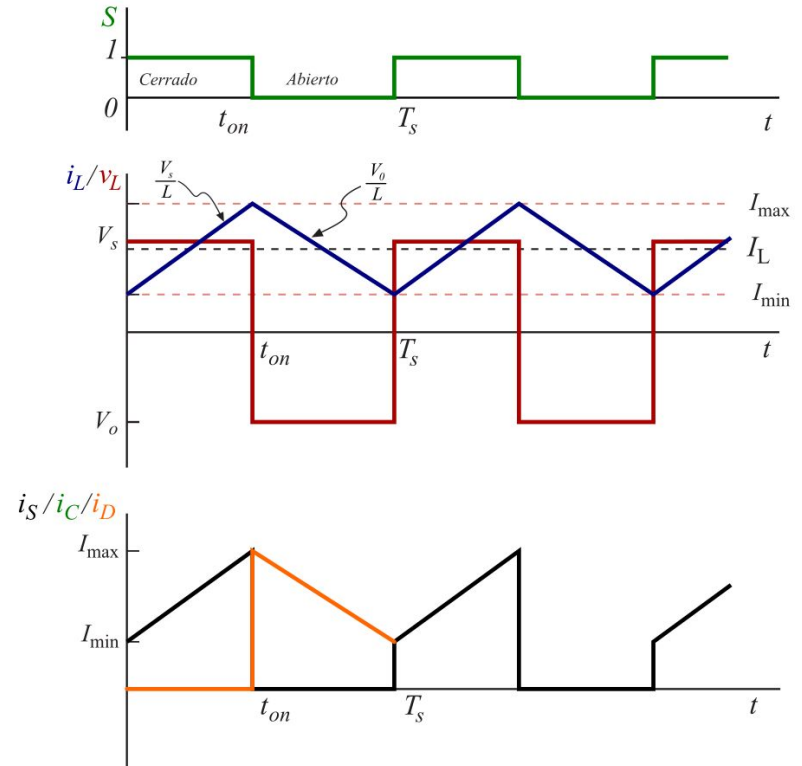
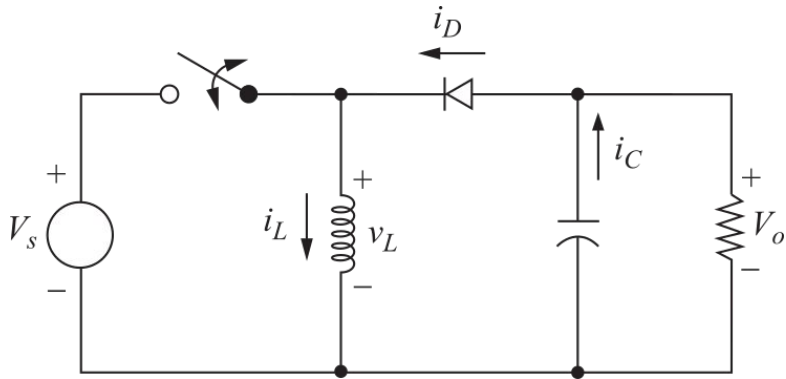
Convertidor Reductor-Elevador: formas de onda



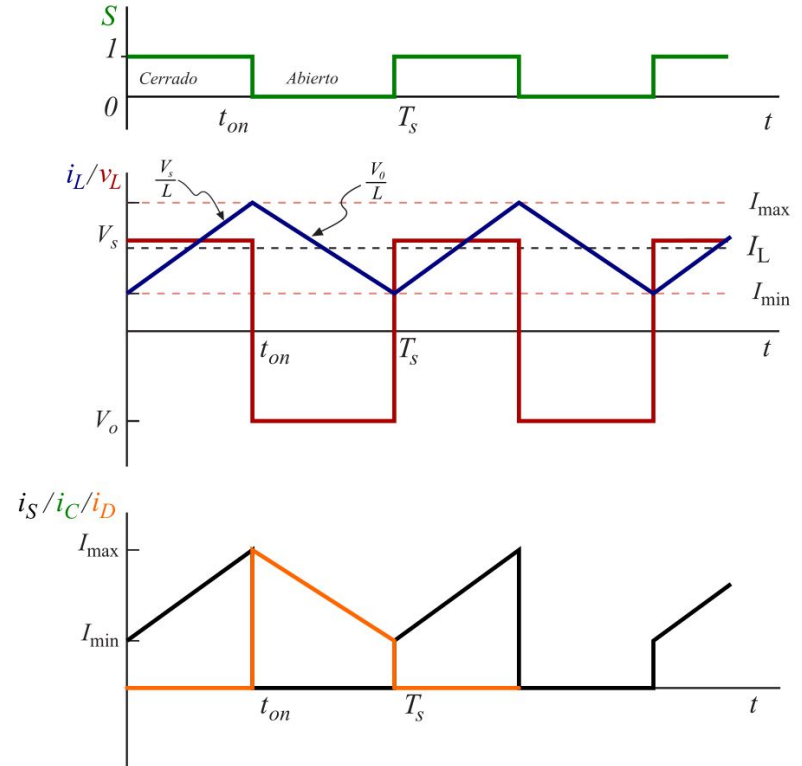
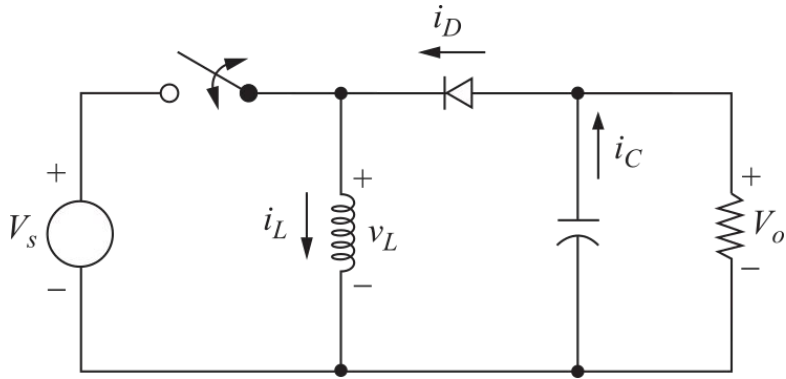
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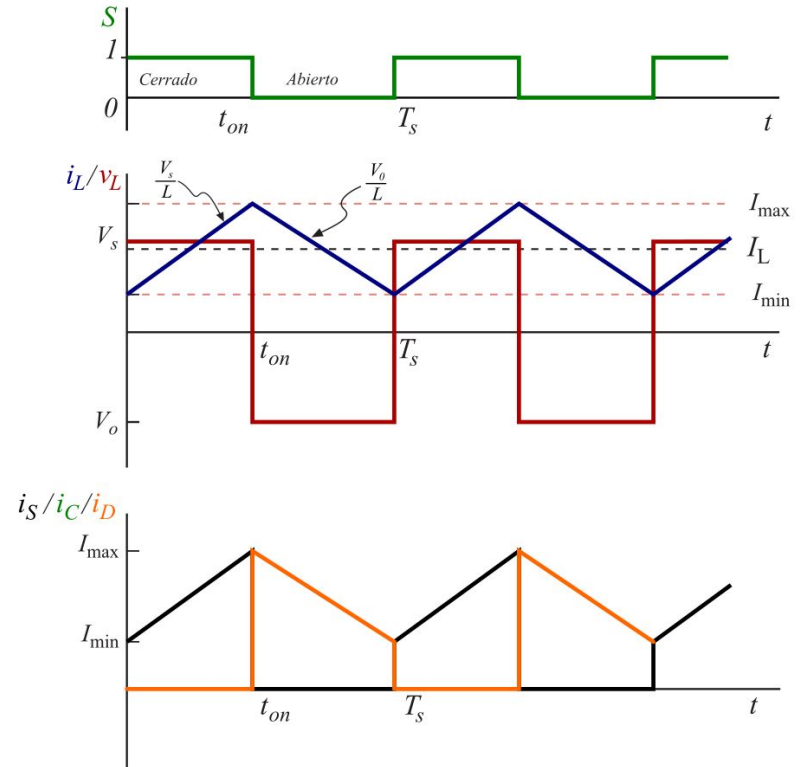
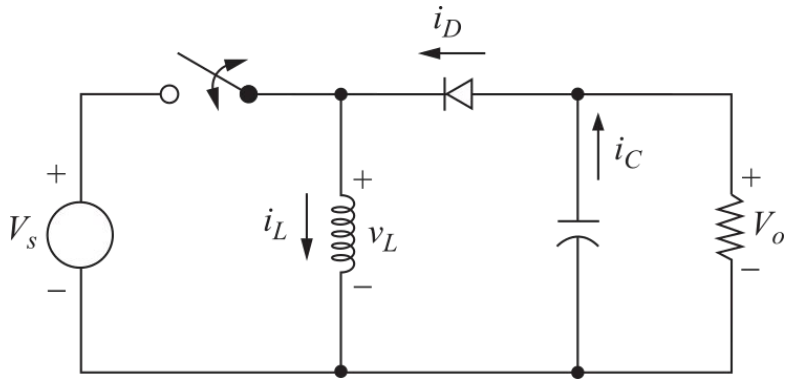
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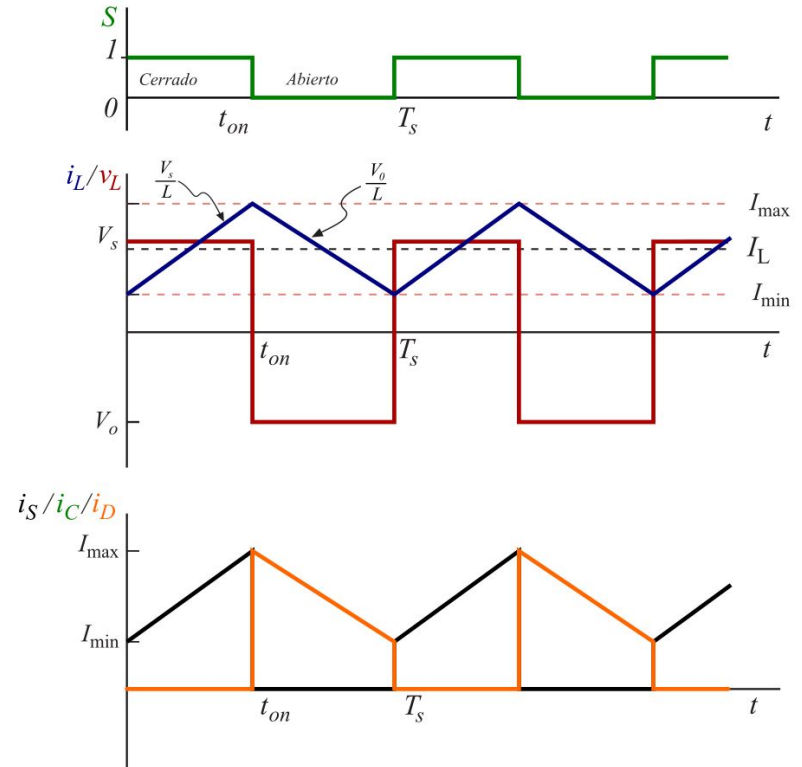
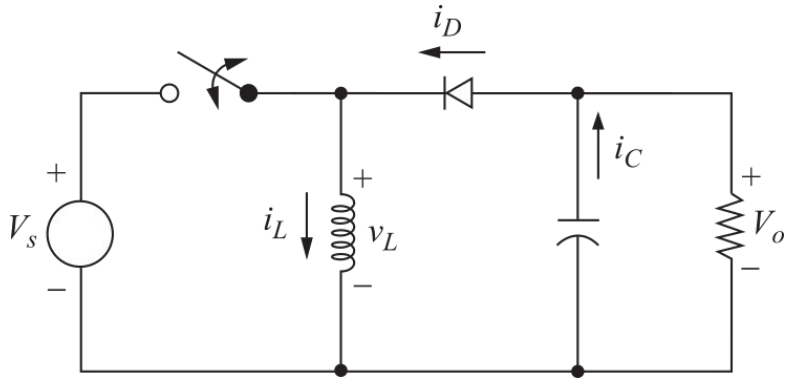
Convertidor Reductor-Elevador: formas de onda



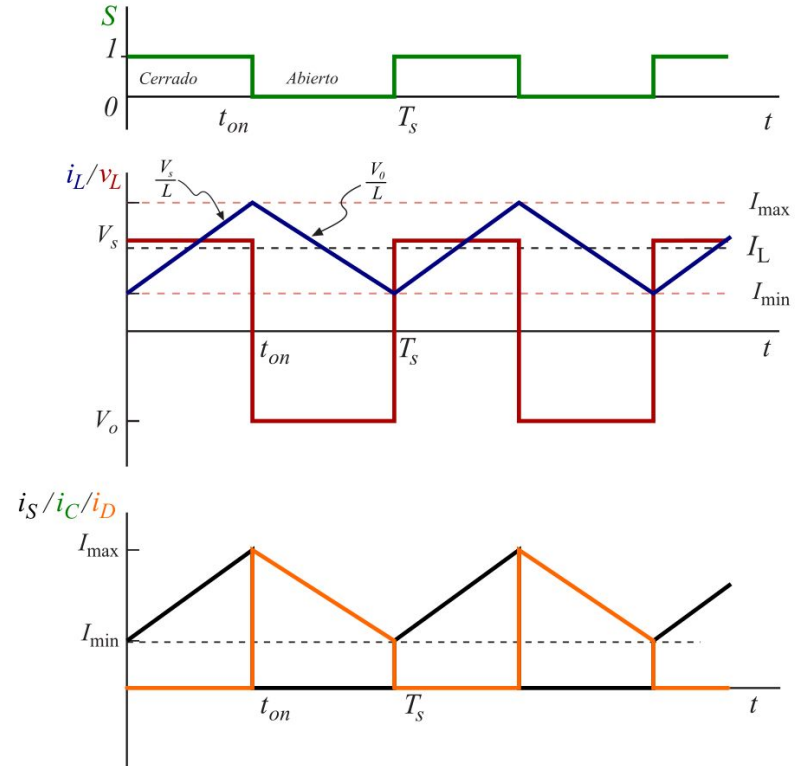
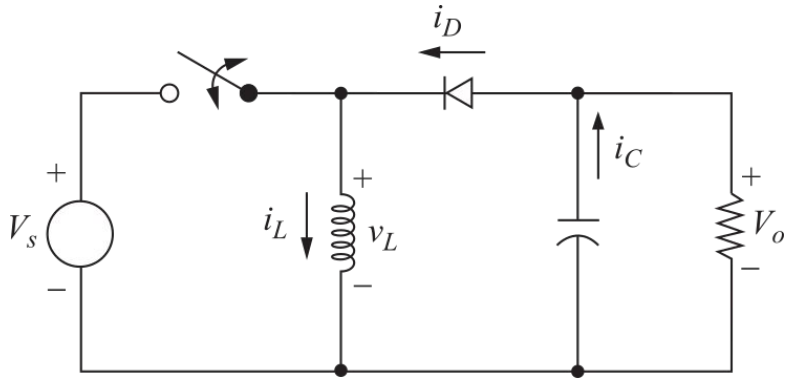
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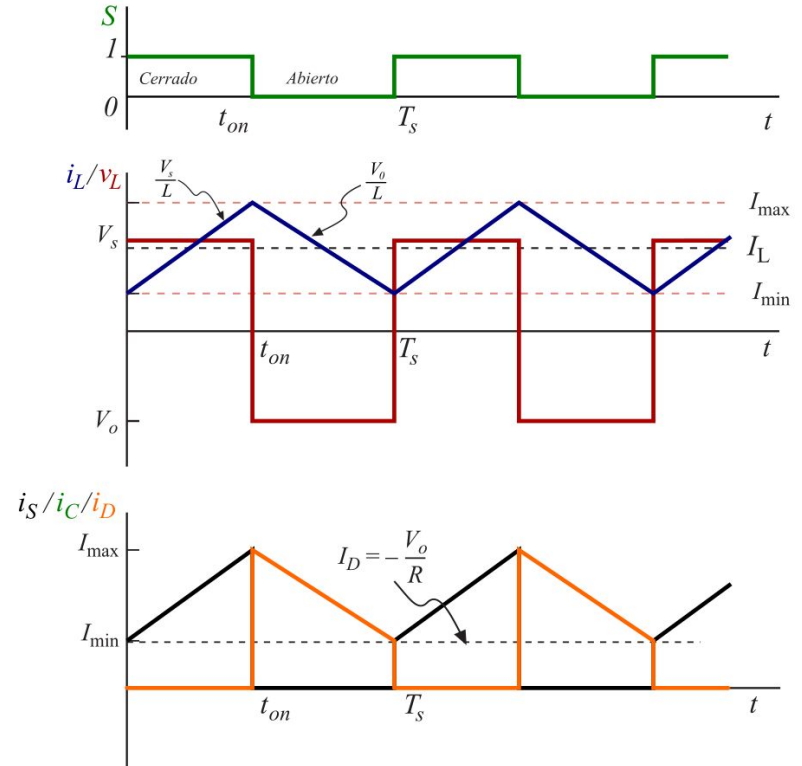
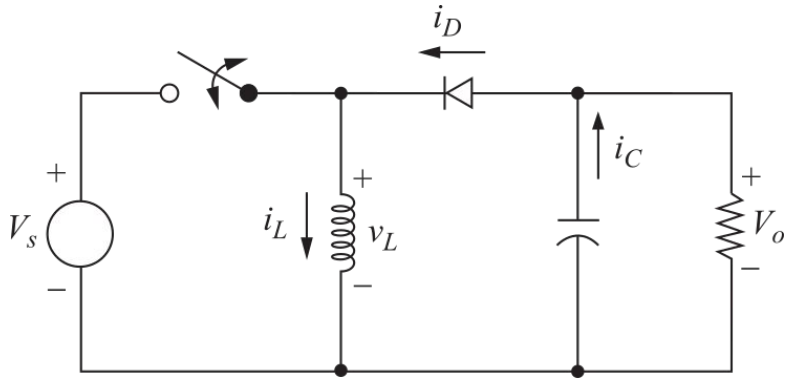
Convertidor Reductor-Elevador: formas de onda



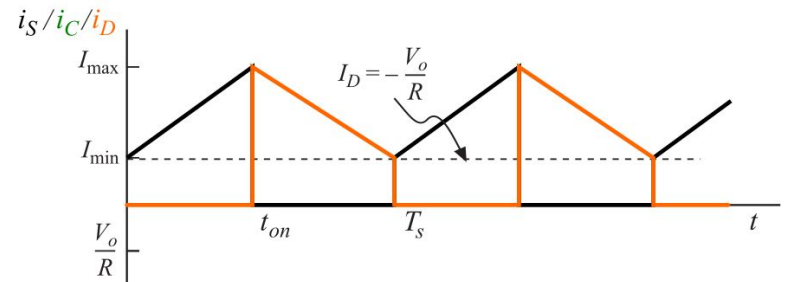
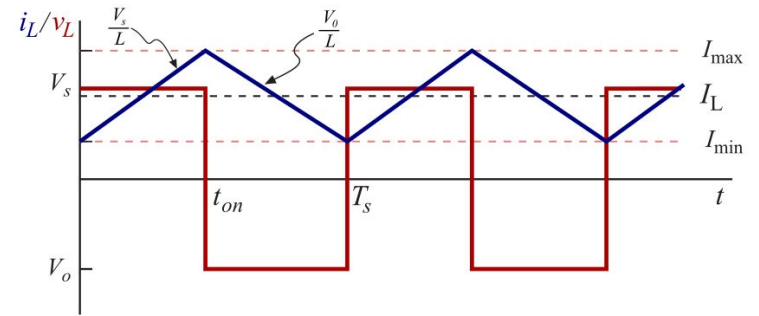
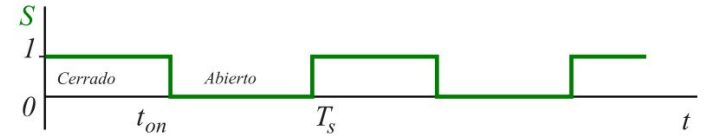
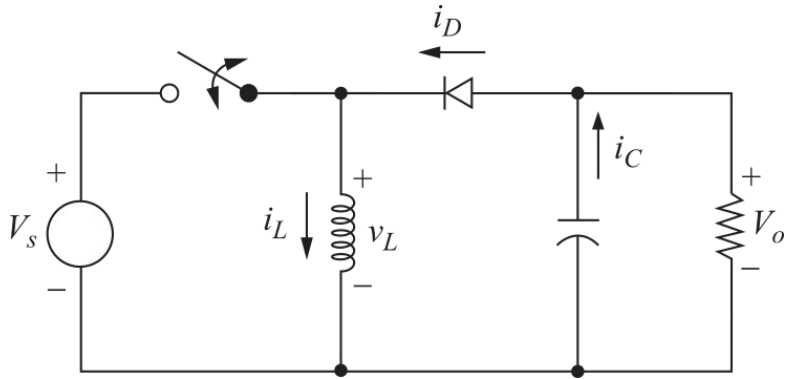
Convertidor Reductor-Elevador: formas de onda



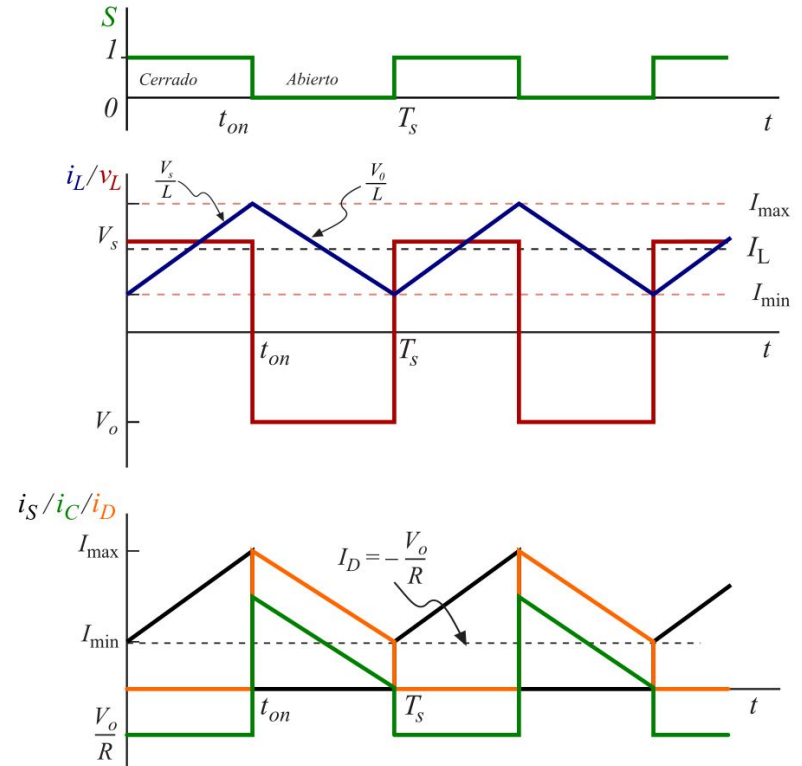
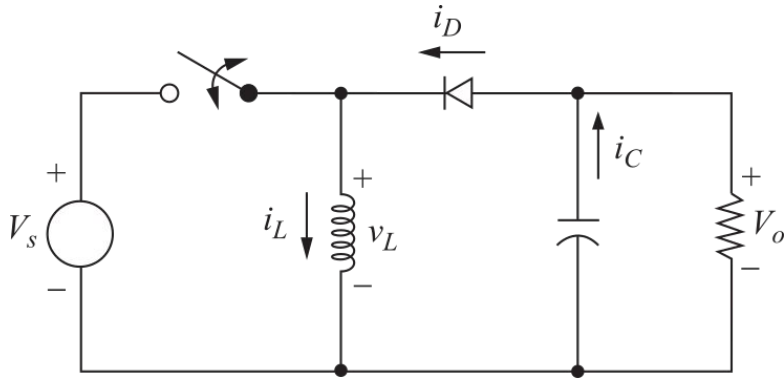
Convertidor Reductor-Elevador: formas de onda



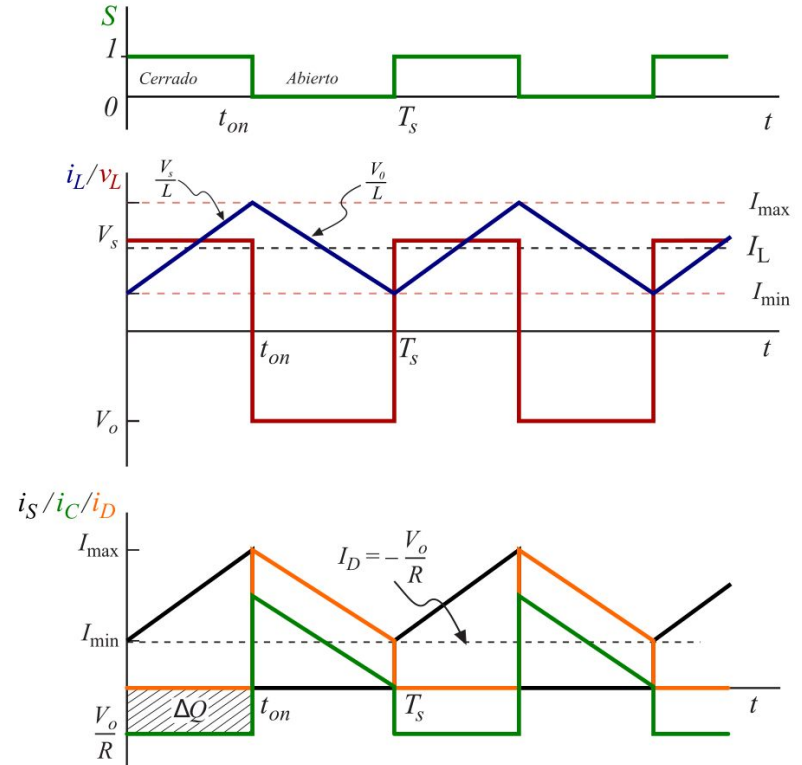
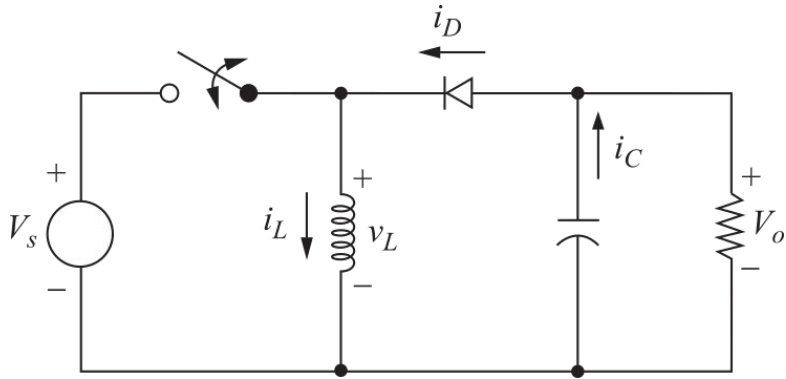
Convertidor Reductor-Elevador: formas de onda



Convertidor Reductor-Elevador: formas de onda

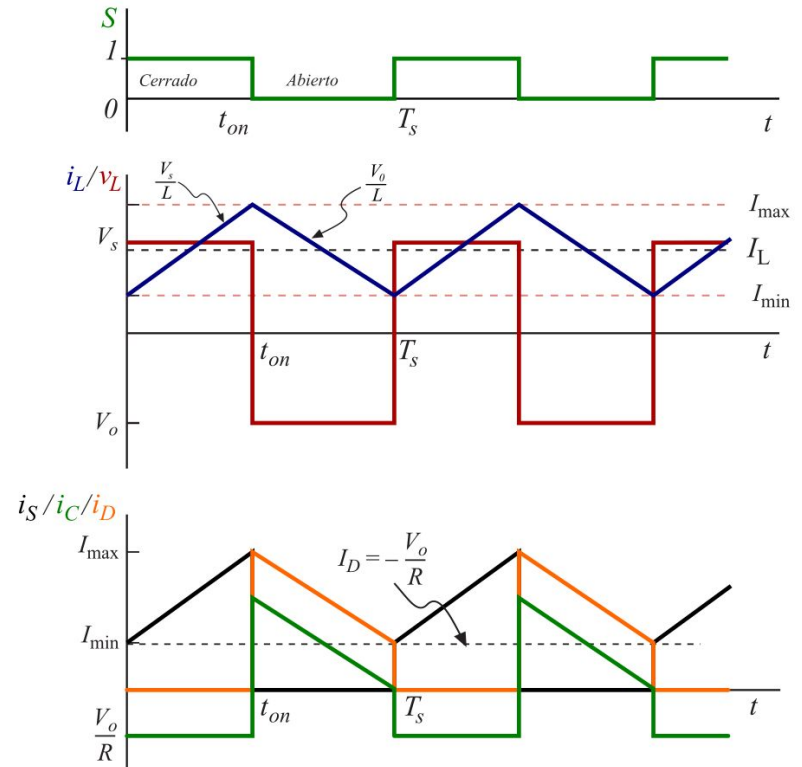


Convertidor Reductor-Elevador: formas de onda



Convertidor Reductor-Elevador: rel. de conversión

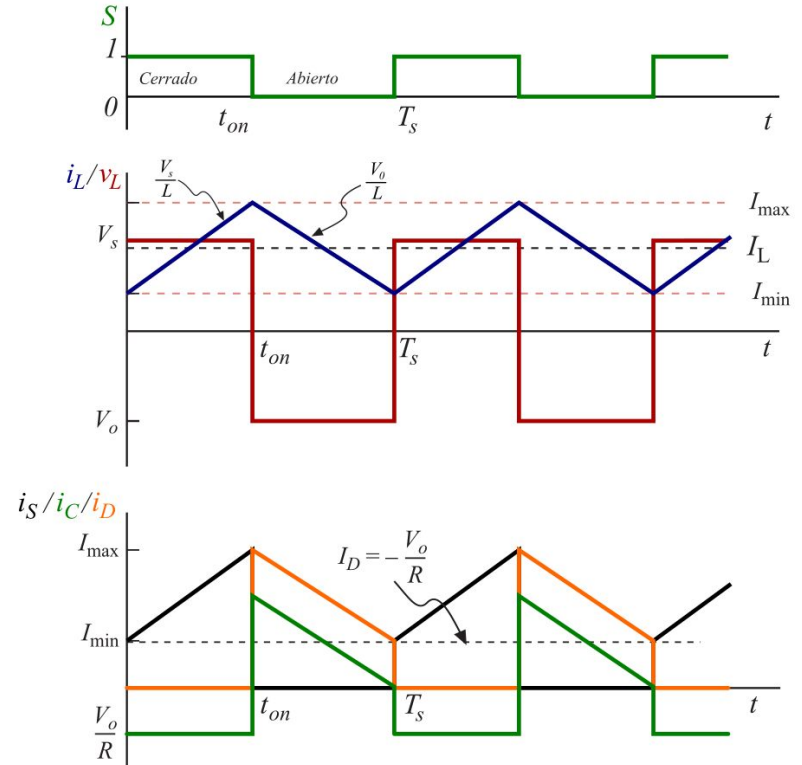
La tensión media en el inductor debe ser cero:



Convertidor Reductor-Elevador: rel. de conversión

La tensión media en el inductor debe ser cero:

$$V_s t_{on} = -V_o (T_s - t_{on})$$

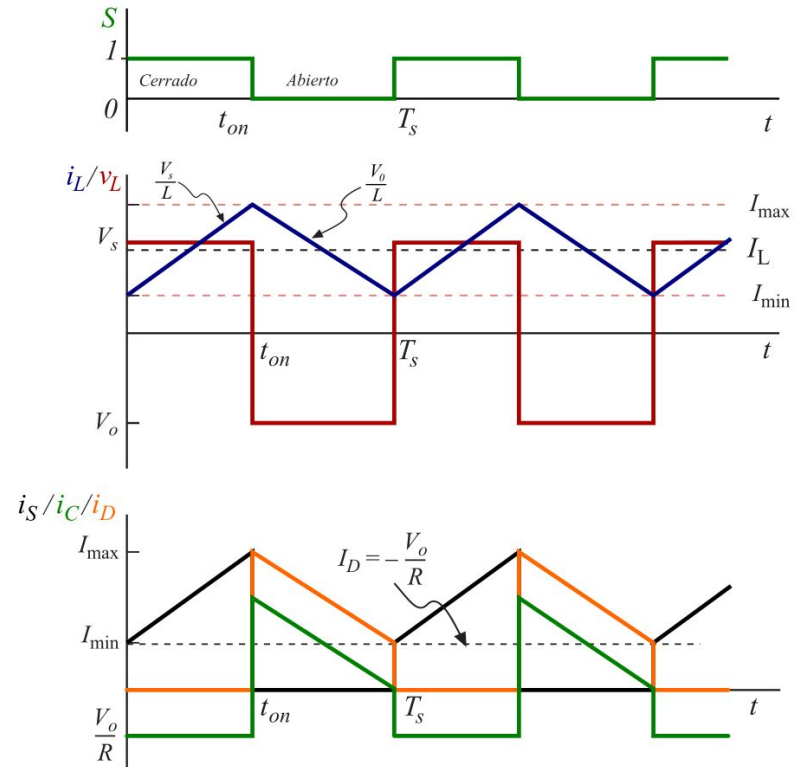


Convertidor Reductor-Elevador: rel. de conversión

La tensión media en el inductor debe ser cero:

$$V_s t_{on} = -V_o (T_s - t_{on})$$

$$V_s \frac{t_{on}}{T_s} = -V_o \left(1 - \frac{t_{on}}{T_s}\right)$$



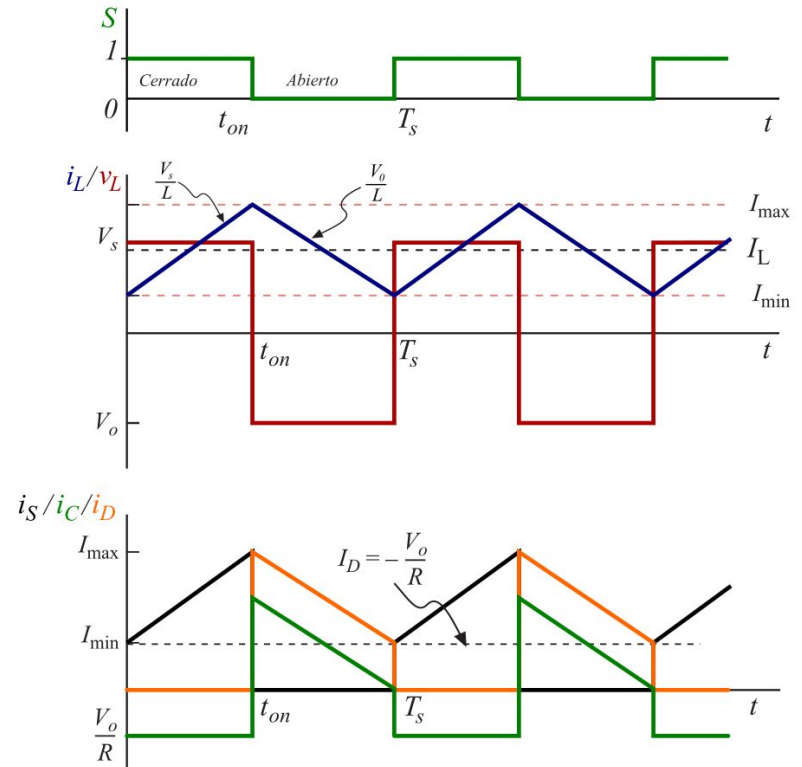
Convertidor Reductor-Elevador: rel. de conversión

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$$V_s D = -V_0 (1 - D)$$



Convertidor Reductor-Elevador: rel. de conversión

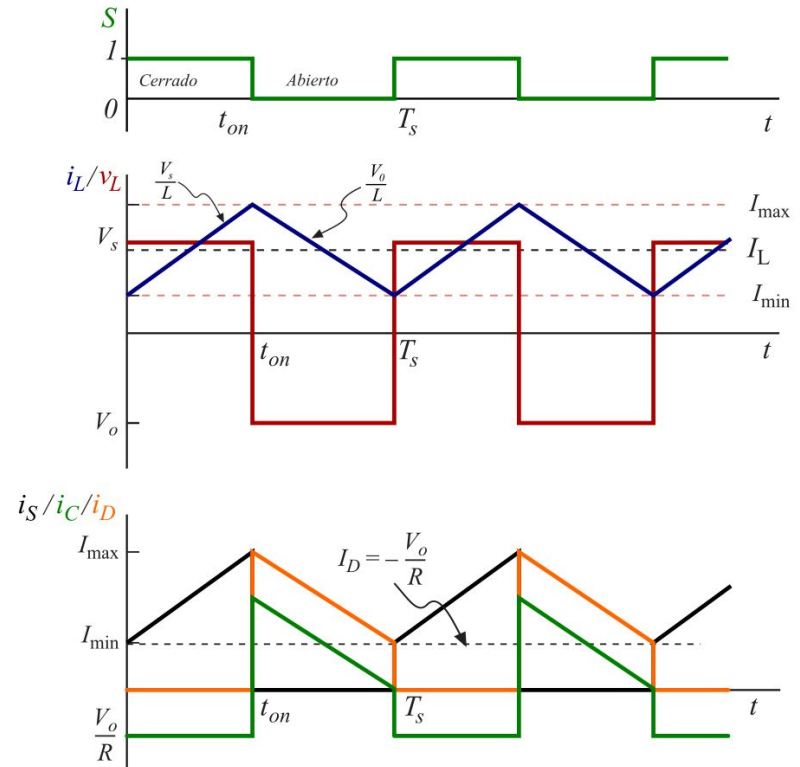
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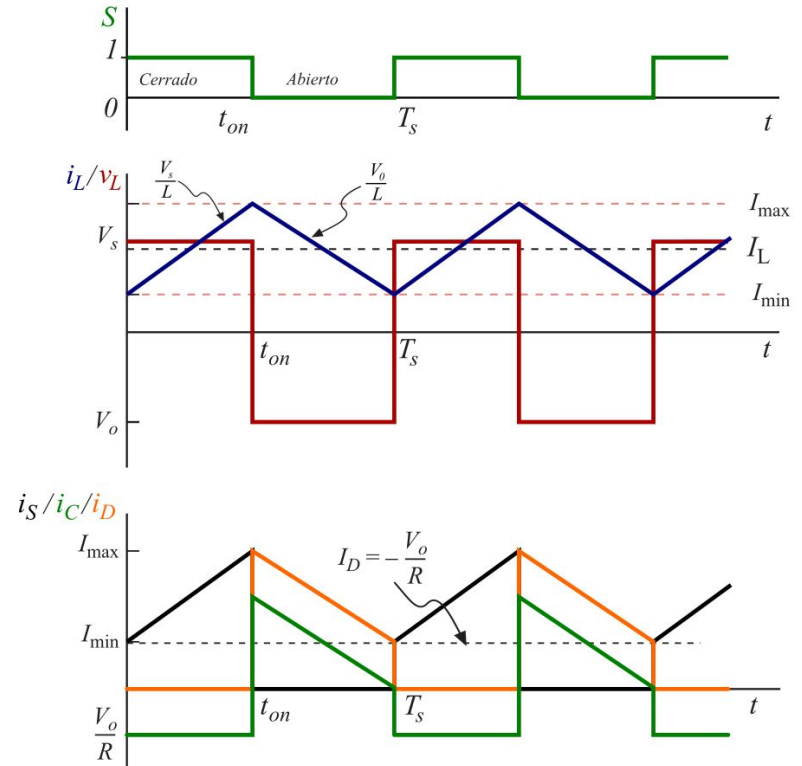
$$V_s D = -V_0 (1 - D)$$

$$V_0 = -\frac{D}{(1 - D)} V_s$$



Convertidor Reductor-Elevador: rel. de conversión

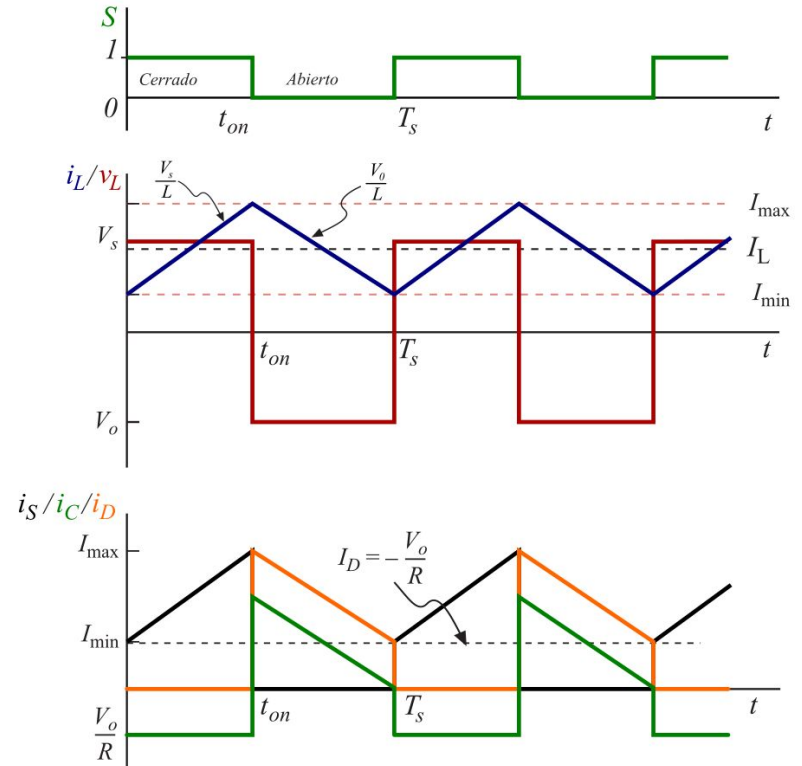
¿Qué pasa con la relación de corrientes?



Convertidor Reductor-Elevador: rel. de conversión

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$$P_s = P_o$$

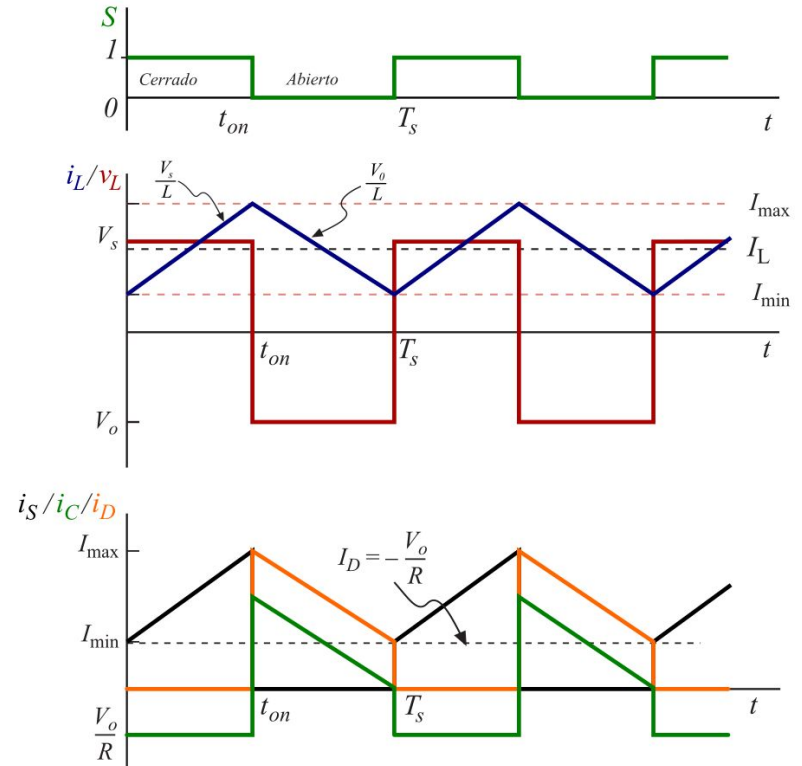


Convertidor Reductor-Elevador: rel. de conversión

¿Qué pasa con la relación de corrientes?

$$P_s = P_o$$

$$\frac{I_s}{I_o} = \frac{V_o}{V_s}$$



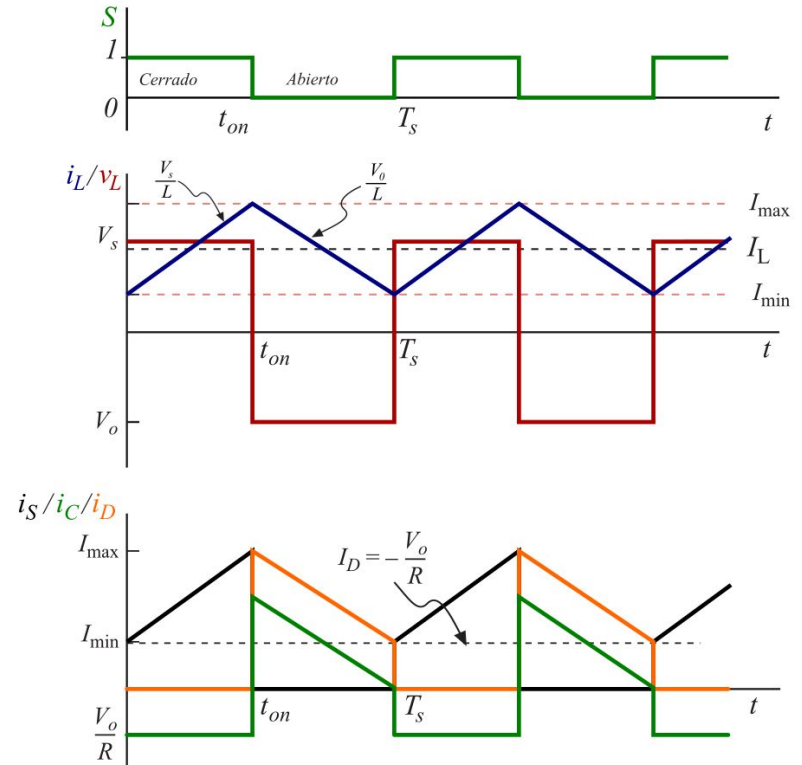
Convertidor Reductor-Elevador: rel. de conversión

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Convertidor Reductor-Elevador: rel. de conversión

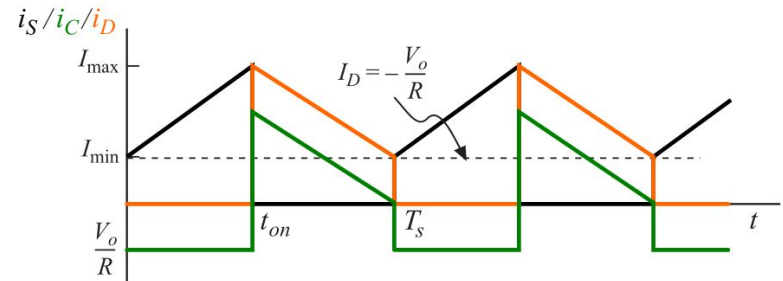
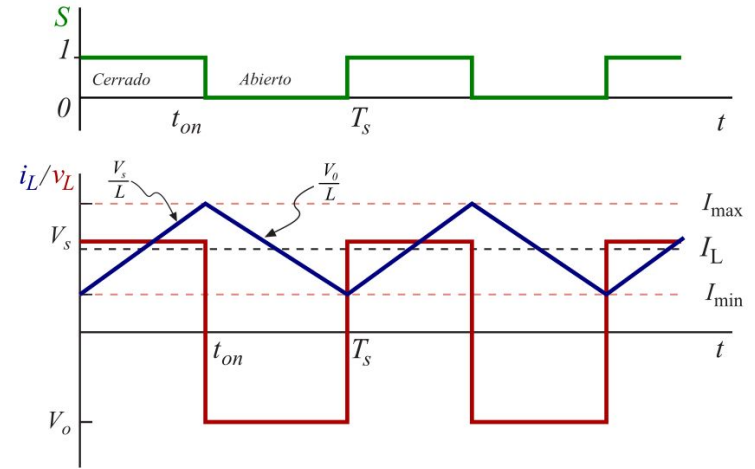
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$$\frac{I_s}{I_o} = \frac{V_o}{V_s}$$

$$V_o = -\frac{D}{(1-D)} V_s$$

$$I_s = -\frac{D}{(1-D)} I_o$$



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